

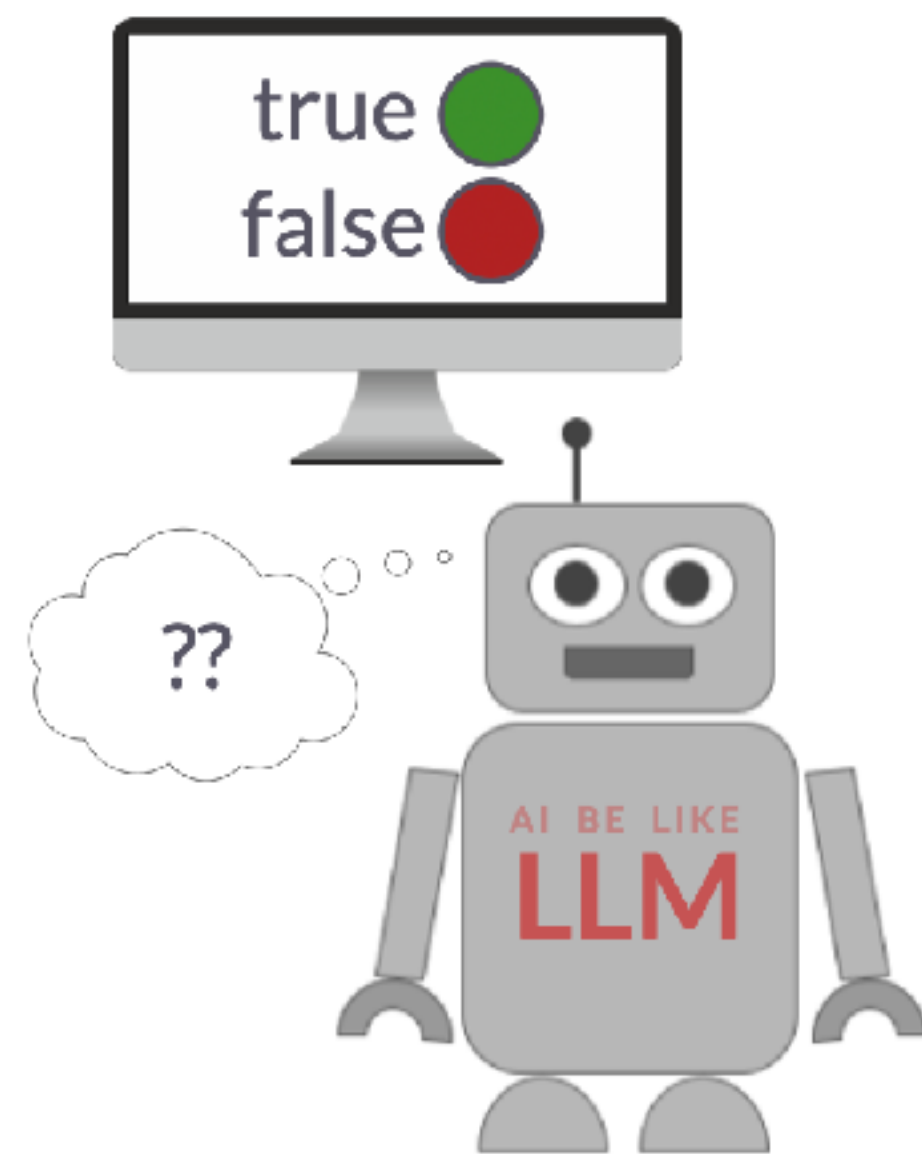
# Understanding Large Language Models

Carsten Eickhoff, Michael Franke and Polina Tsvilodub

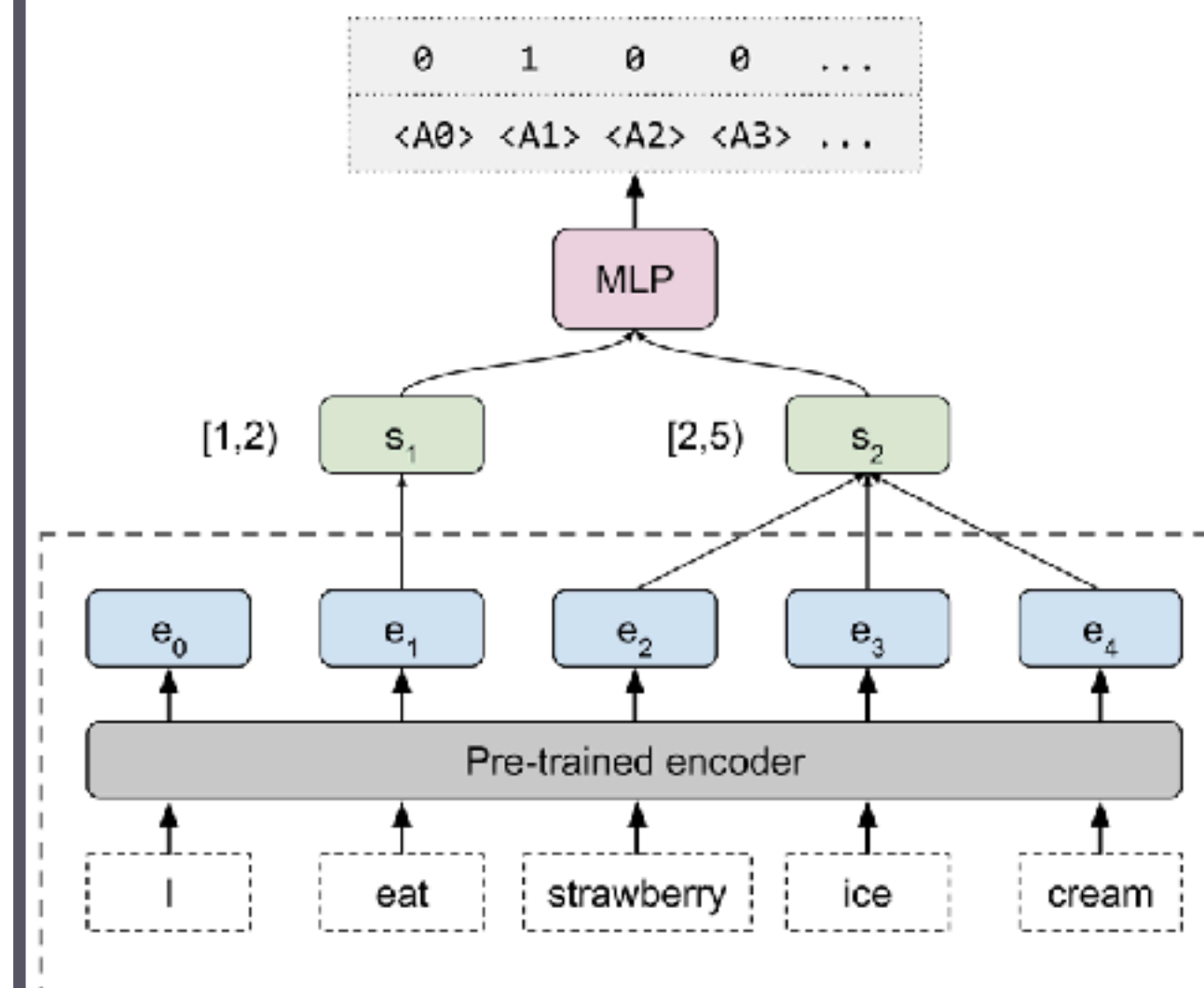
**Session 09: LMs meet Linguistics & Cognitive Science**

# Assessing what LLMs “know” and “can do”

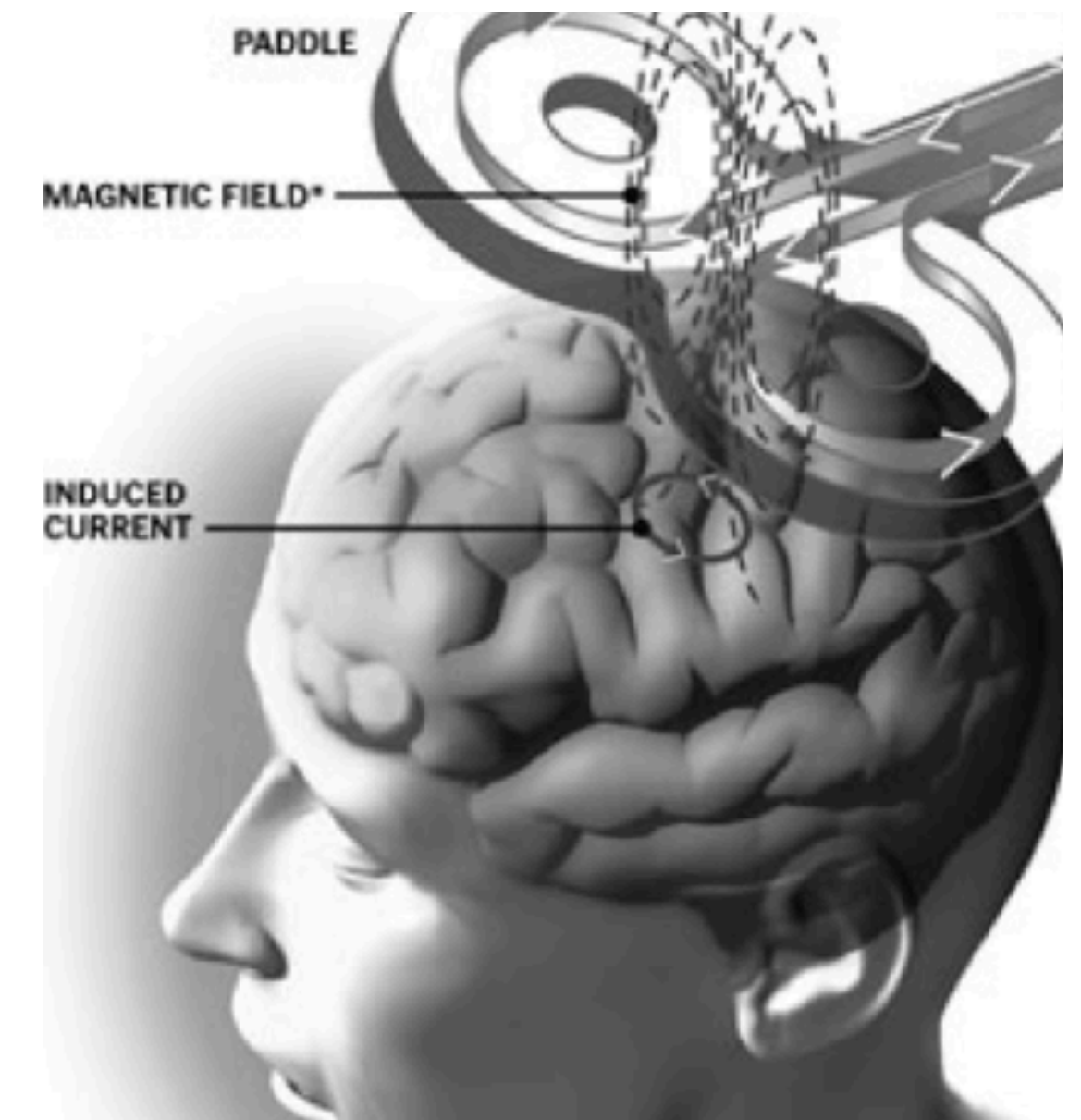
## Behavior. Tests



## Probing



## Intervention



# Main learning goals

## 1. behavioral assessment types & motivation

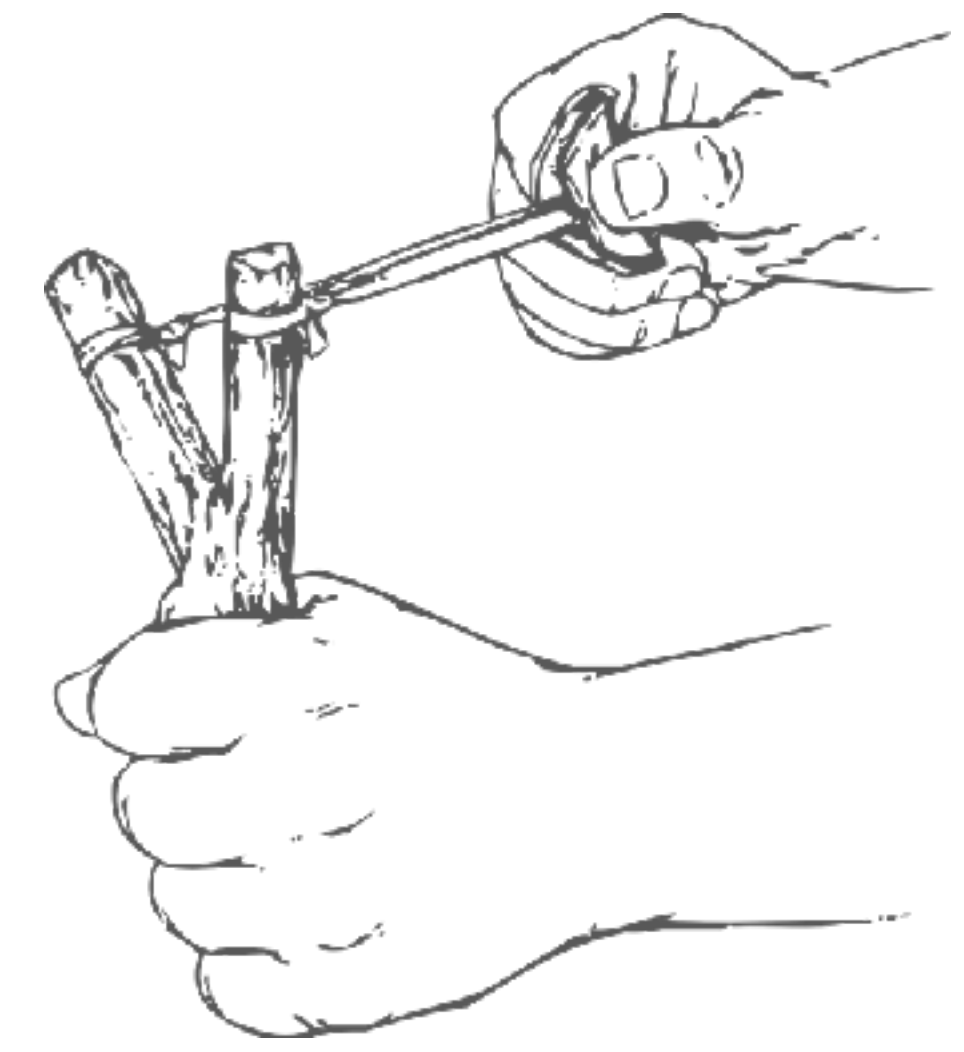
- “common-sense axiom” of behavioral assessment
- accuracy, bias correction, calibration

## 2. machine psychology & LM-human comparison

- LM behavior vs. processing
- targeted assessment

## 3. LMs & computational psycholinguistics

- sentence processing & surprisal theory

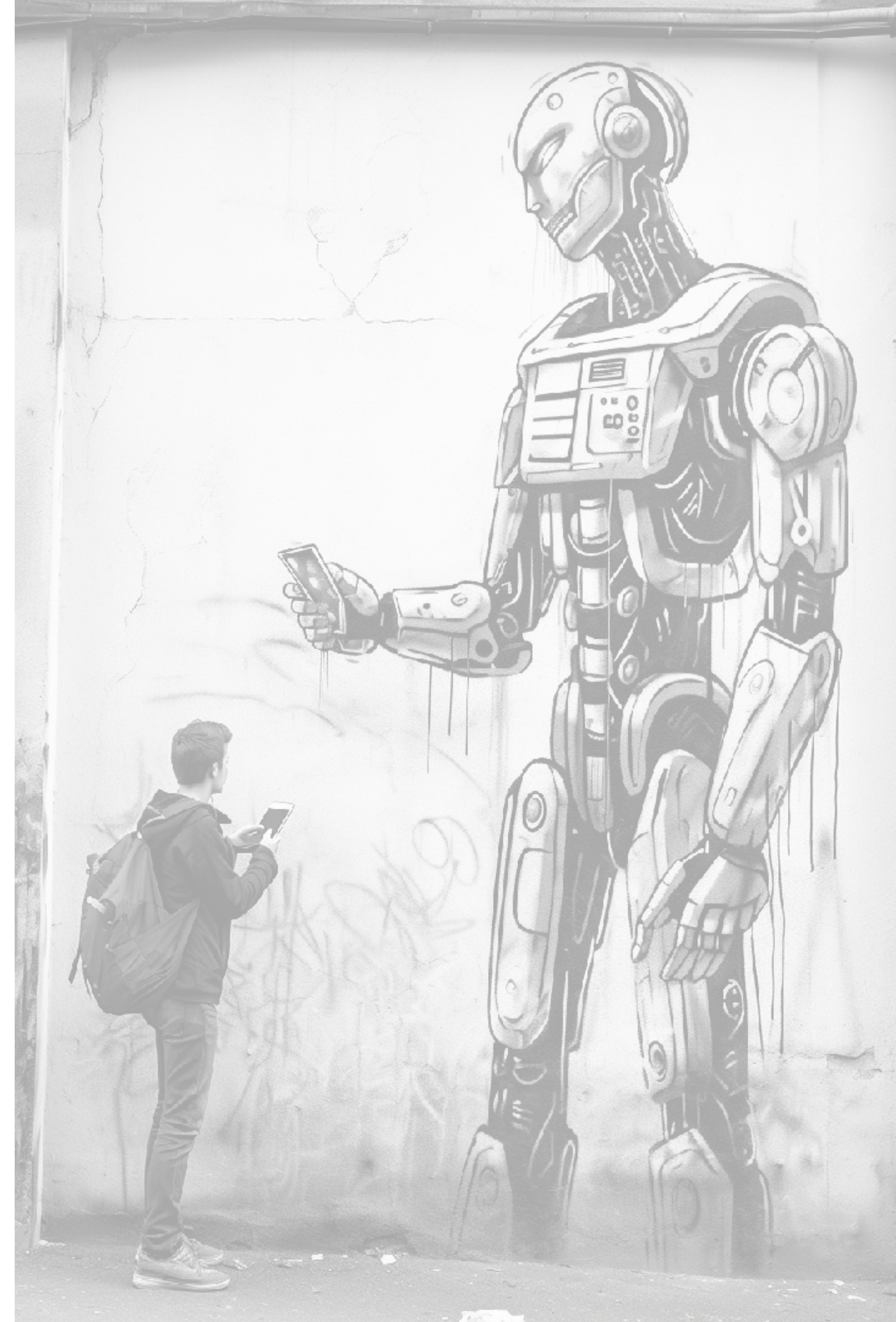




# Behavioral assessment: Motivation

Assessing the behavior of a trained LM

- ▶ is the I/O behavior in particular **applications**:
  - correct
  - trustworthy
  - reliable
  - robust
  - aligned with human expectation / values
  - ...
- ▶ is the I/O behavior **human like**:
  - correct when humans are correct
  - incorrect when humans are incorrect
  - of the same type as what humans do
  - ...



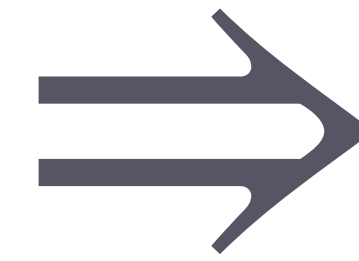


# Behavioral assessment: Motivation

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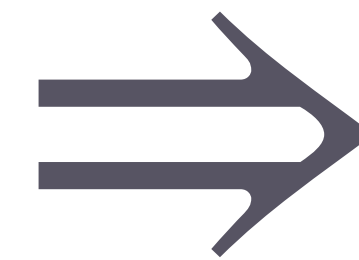


► good applications

- correct
- safe
- ...

► is the I/O behavior **human like**:

- correct when humans are correct
- incorrect when humans are incorrect
- of the same type as what humans do
- ...



- when/ how to anthropomorphize?
- understand how system works
- shed light on human by comparison to machine cognition

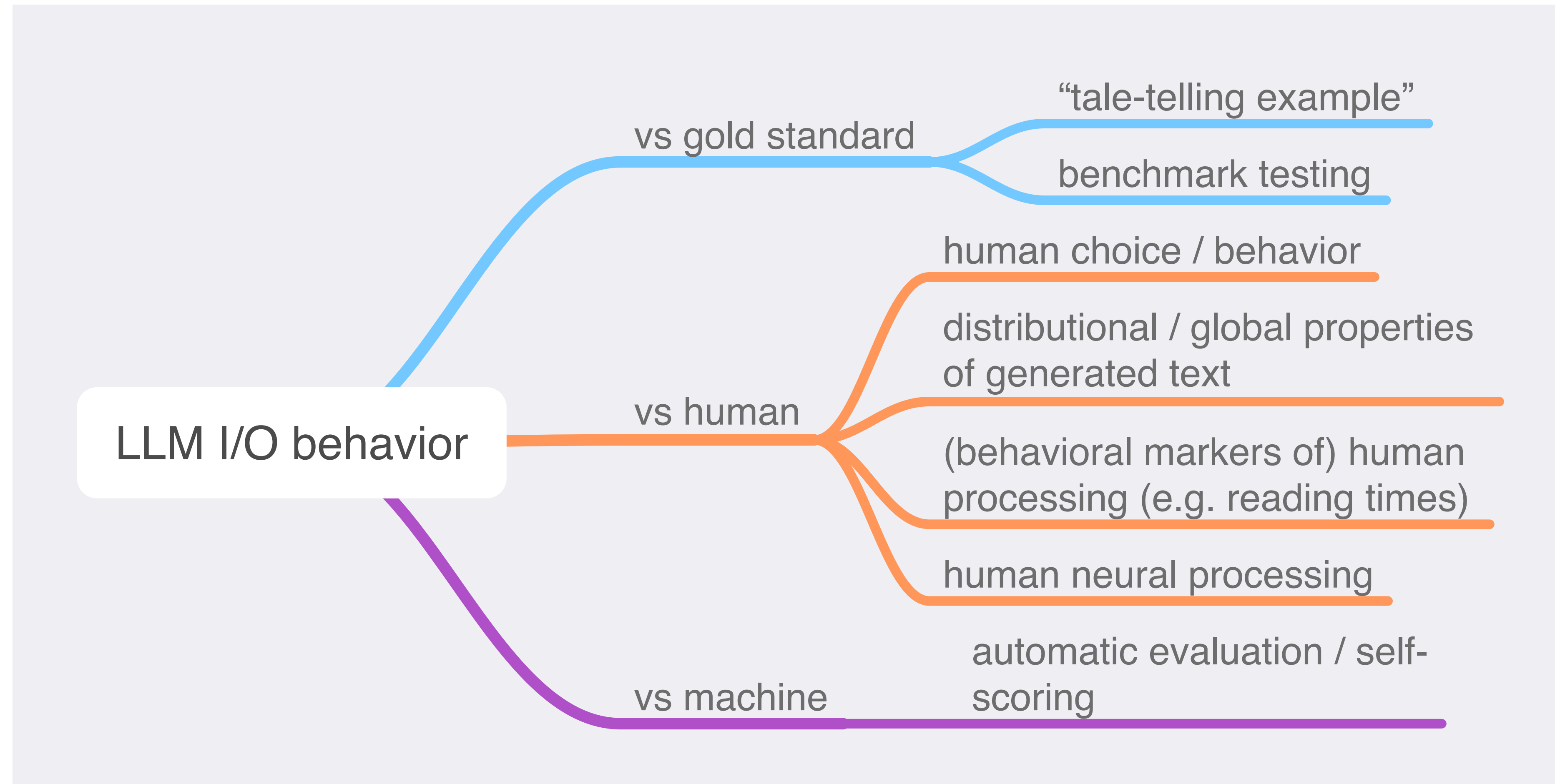
# “Common sense axiom” of behavioral assessment

for downstream applications

If we want to assess the behavior of system  $S$  because we want to know how well  $S$  performs in an application context  $C$ , then we should test the behavior of  $S$  under the exact same circumstances as  $S$  would be used in the application context  $C$ .



# Overview





**“Tale-telling” by example**

# Motivating examples

in research papers

by different individuals. LMs additionally exhibit some success in relating beliefs to plans:

*Lou leaves a lunchbox in the work freezer. Syd, who forgot to pack lunch, eats Lou's instead. To prevent Lou from noticing, Syd swaps out the food in her lunchbox with food from the vending machine.*

This completion that describes multiple individuals' differing motivations and beliefs along with information about the environment in which they act.

What these examples suggest, and what I want to argue in the rest of this paper, is that LMs can serve as **models of agents** in a narrow sense: they can predict relations between agents' observations, internal states, and actions or utterances. In partic-

← single prompt completion

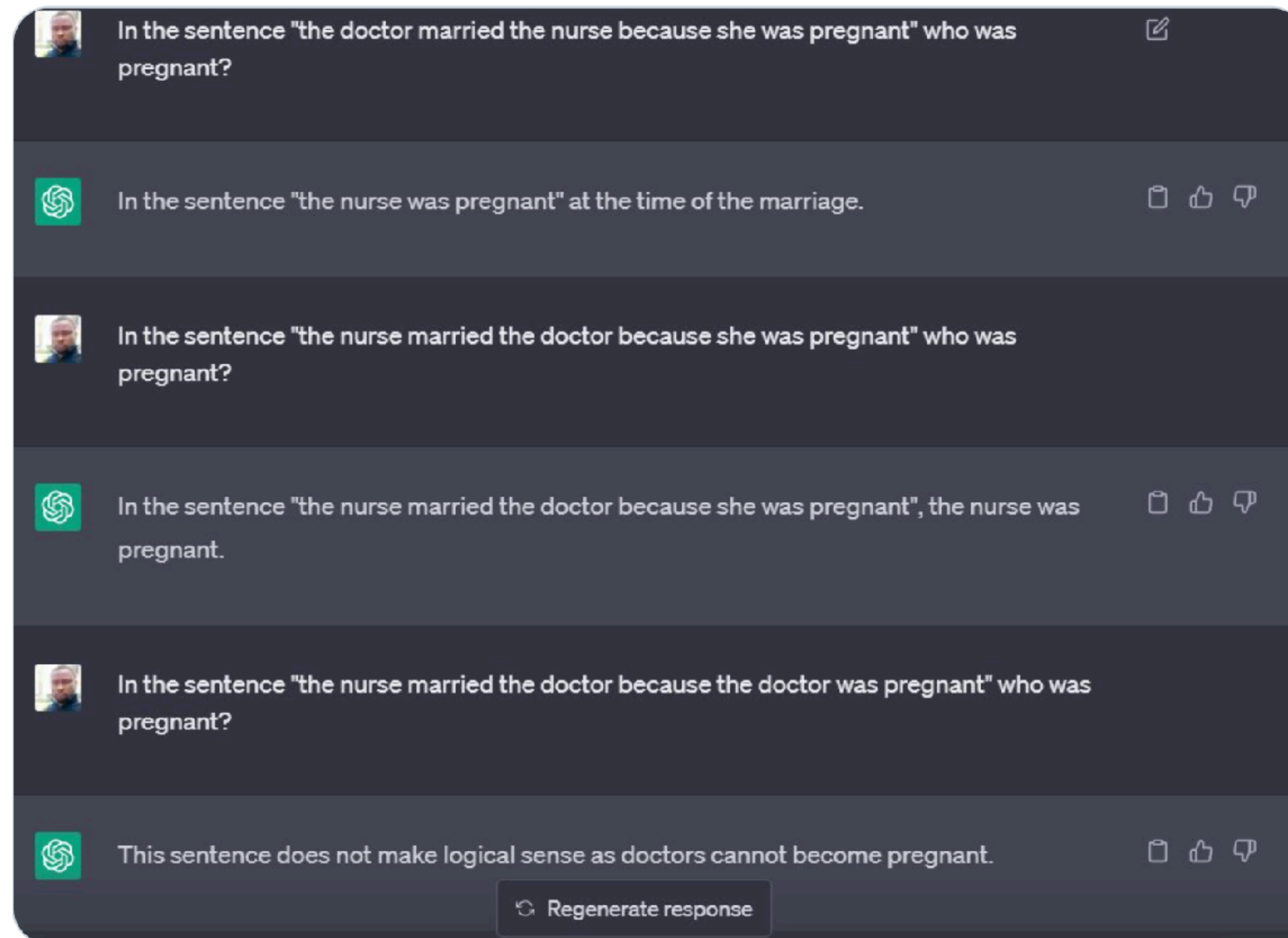
← interpretation

← serves as motivation for more in-depth analysis



# Critical failures (e.g., ethical considerations)

on social media



# Single examples as key arguments

in scientific writing

## 5. Likely vs. Grammatical

LLMs also lack a distinction between likelihood and grammaticality. The two notions often overlap, but they are conceptually distinct: some things are unlikely but correct, and others are likely but incorrect. Human speakers can tease these notions apart (as Chomsky, 1957 pointed out and as discussed in further detail by Berwick, 2018; Sprouse et al., 2018, and others). LLMs cannot: any attempt they might make to judge goodness is based on likelihood.<sup>7</sup> This means that they will generally prefer an ungrammatical but somewhat likely continuation over a grammatical but unlikely one:

**User:**

Here is the beginning of an English sentence: "The little duck that met the horses with the blue spots yesterday ...". Which of the following two words is a better next word, and why? (a) are, (b) destroys.

**GPT:**

The better next word in this case would be (a) "are". The sentence is describing a situation in which the little duck met the horses, so it makes sense to continue with a verb that maintains the context. "Are" is a plural form of "to be" and can be used to describe the state of the horses or a characteristic they possess. In this case, it could be used to further describe the horses with the blue spots.

Option (b) "destroys" changes the context abruptly and introduces a violent action that is not in line with the rest of the sentence. It would not fit well within the context provided by the beginning of the sentence.

In this case, ChatGPT chose the ungrammatical but frequent word "are" as the continuation instead of the grammatical (but perhaps unlikely) continuation "destroys". Future models may of course perform better on examples such as the above, but the point remains: the distinction between likely and grammatical, which all humans have, is entirely foreign to ChatGPT and its fellow LLMs.

## 6. Generalization

Taking a step back from the linguistic behavior of LLMs, let us look at how such models

← very general issue

← single example from a single model with a single method of assessment

← strong conclusion

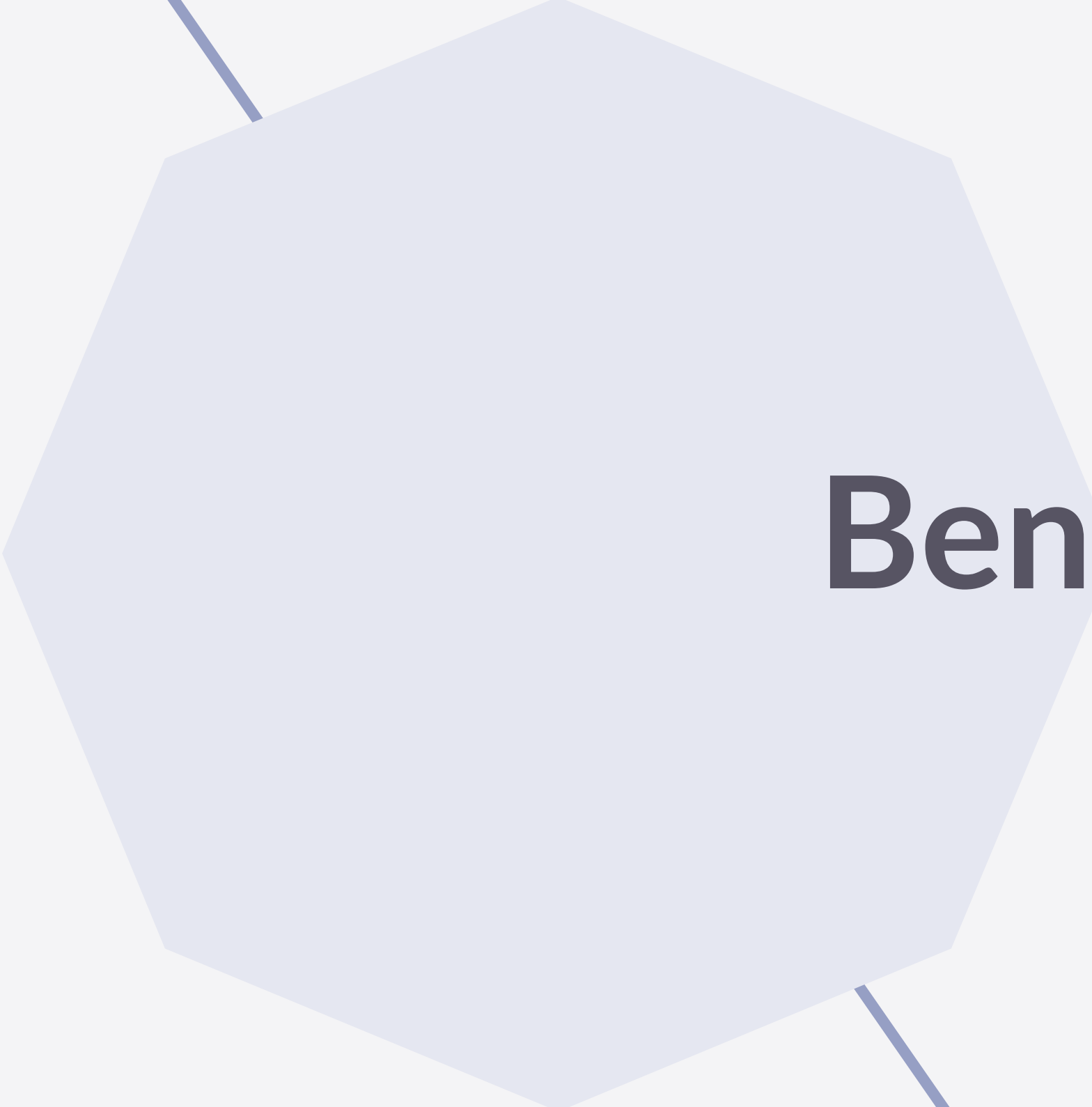


# Taking stock: “tale-telling” by example

- ▶ good for motivation or suggestive demonstration
- ▶ strong conclusions only if
  - a single example is evidence enough
    - e.g., to refute a universal statement
    - e.g., system must not get a single instance wrong ever
  - back-up by more systematic testing
    - several items
    - several models
    - different assessment methods
    - ideally: **antagonistic testing**







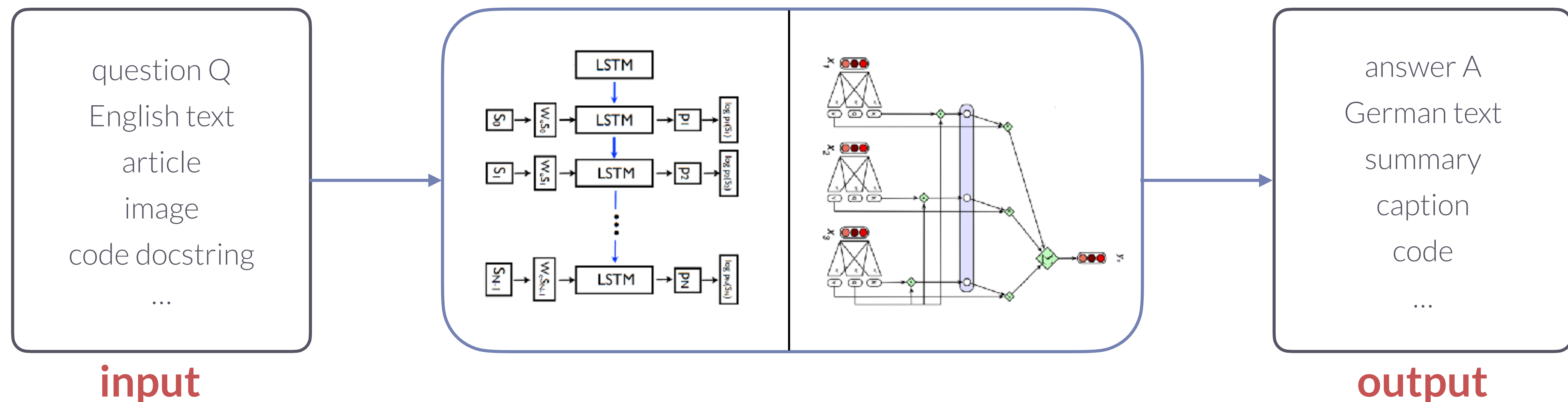
**Benchmarks**

## ► what is it?

- standardized **set of tasks** or datasets with fixed **standards of evaluation**
- designed to evaluate performance & capabilities

## ► what is it **for**?

- systematic assessment of (emergent) capability
- streamlined comparison of different systems
  - e.g., **scaling** (performance boost from size *ceteris paribus*)
- track advancement in the field
- identify strengths and weaknesses
- feedback for future work
- define “what the community cares about”



# Example: Causal Judgement

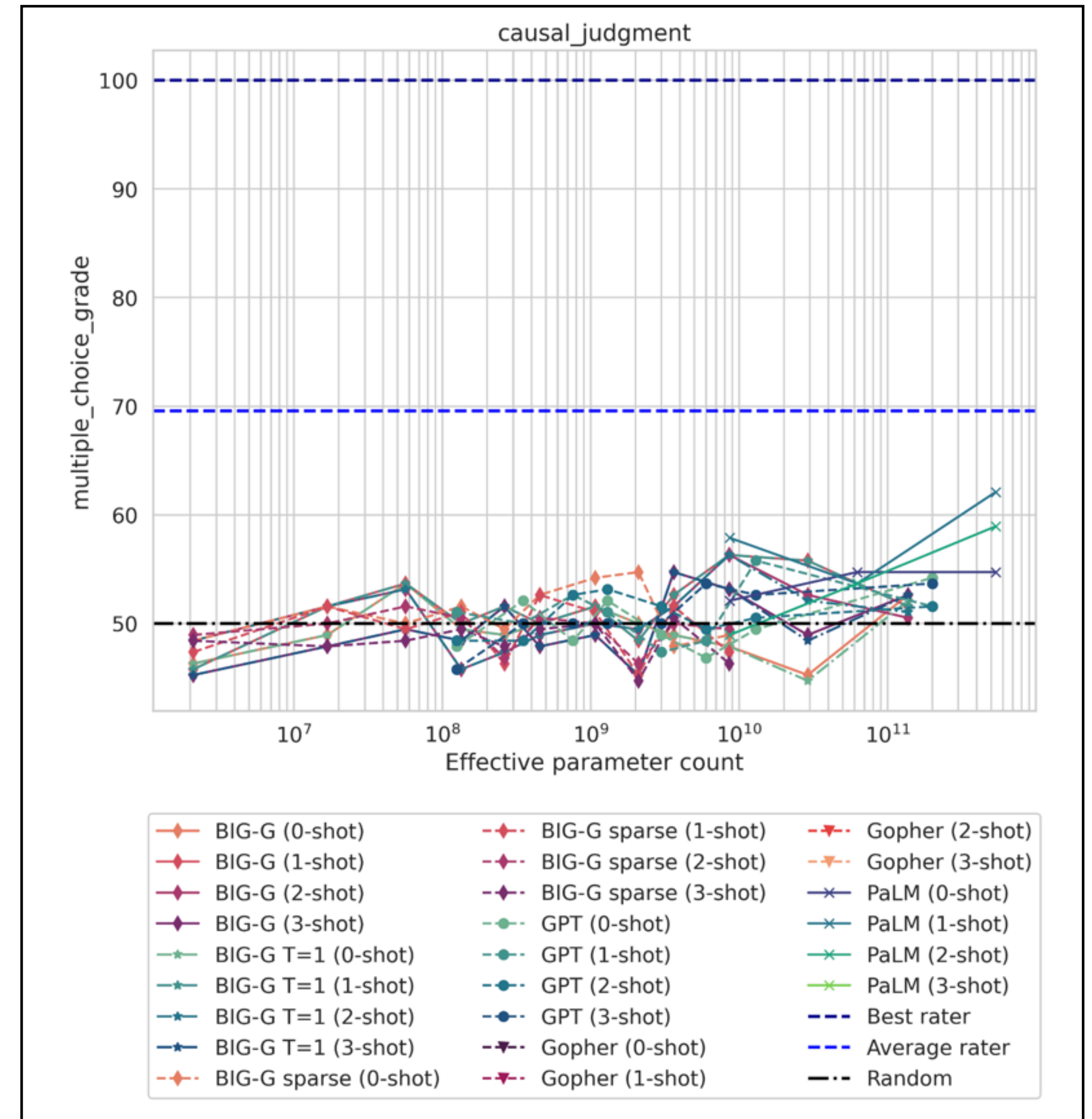
part of BIG-Bench

How would a typical person answer each of the following questions about causation?

The CEO of a company is sitting in his office when his Vice President of R&D comes in and says, 'We are thinking of starting a new programme. It will help us increase profits, but it will also harm the environment.' The CEO responds that he doesn't care about harming the environment and just wants to make as much profit as possible. The programme is carried out, profits are made and the environment is harmed. Did the CEO intentionally harm the environment?

☒ yes

☐ no





# Example: Empirical Judgements

part of BIG-Bench

Determine whether a given sentence asserts a causal, correlative, or neutral relation between two events. If the sentence asserts a causal relation respond causal, if the sentence asserts a correlative relation respond correlative, if the sentence asserts neither a causal nor a correlative relation between two events respond neutral.

If the sun shines on the stone, the stone becomes warm.

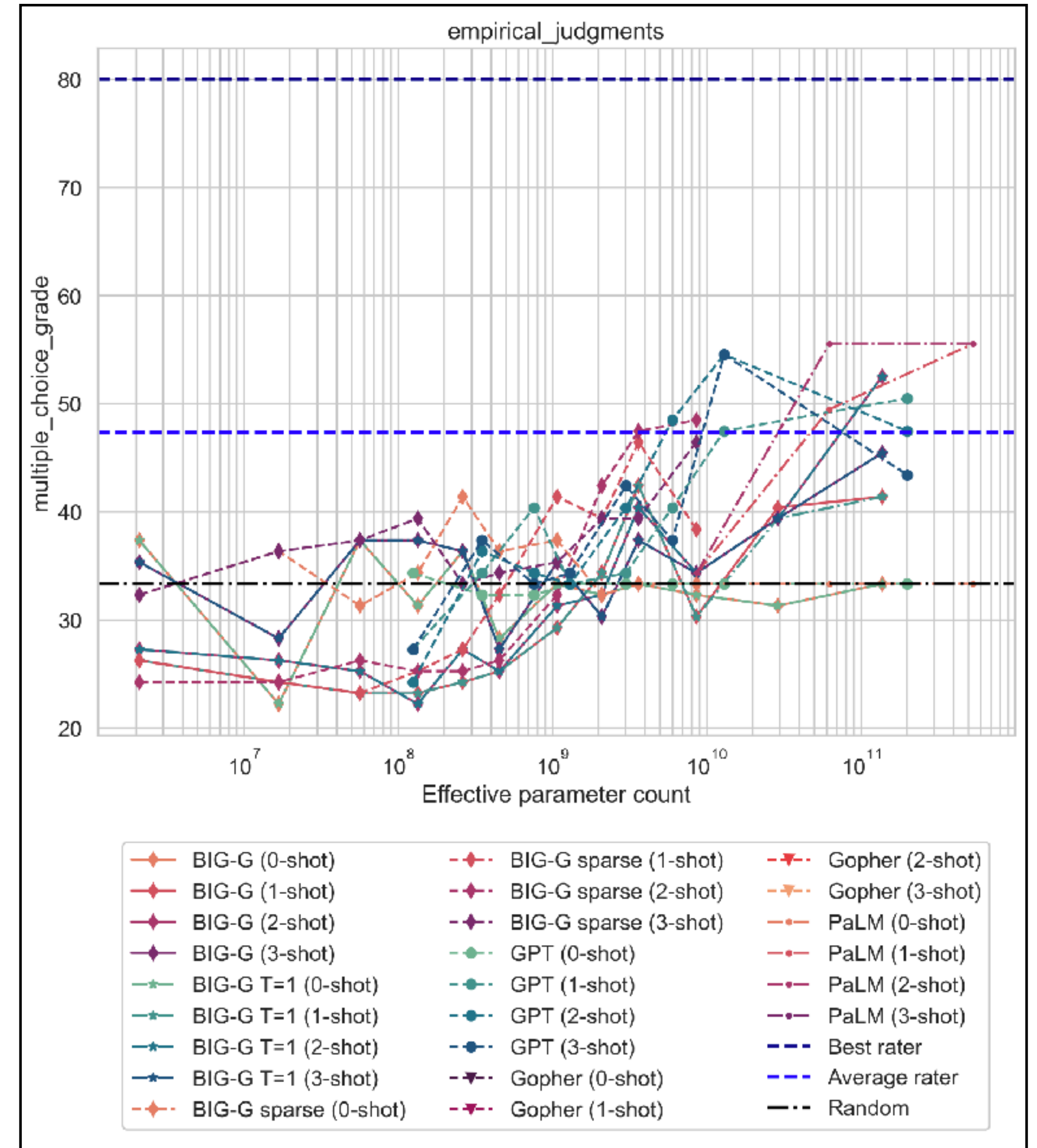
☐ causal      ☒ correlative      ☐ neutral

The sun warms the stone.

☒ causal      ☐ correlative      ☐ neutral

If Boo is a cat, Boo is a feline.

☐ causal      ☐ correlative      ☒ neutral





# Think break

Put on your fencing gloves and try to as critical as possible about the last two examples of benchmark tests. Ask yourself, for example:

- ▶ How informative, in general, is it to learn how well some agent or model performs on the task?
- ▶ Is task performance closely linked to whatever the task description says the task is testing for?
- ▶ Are the task items well designed?
  - material can be accessed [here](#) and [here](#)
- ▶ If you know how an agent performs on this task, do you learn about whether the agent “understands” the relevant concepts addressed by the task?





# Taking stock: benchmark testing

- ▶ large scale assessment of system capabilities
- ▶ batteries of standardized tests
- ▶ systematics comparison (e.g., scale effects)
- ▶ sets 'common research agenda'
- ▶ individual tests can be of poor-ish quality
  - benchmark test called "X" is not necessarily carefully designed to test for ability "X"
- ▶ **creating informative benchmarks is non-trivial**
  - requires awareness of proper experimental methods







**Evaluation metrics  
when a gold-standard exists**

# Accuracy (standard winner-takes all version)

in multiple-choice / classification tasks

- ▶ set of items  $I$
- ▶ each item  $i \in I$  consists of
  - $x_i$  input prompt (string)
  - $y_{i1}, \dots, y_{ik}$  set of answer options (strings)
  - designated “true” option  $t_i$
- ▶ **raw score** for each option:
$$s(y_{ij}) = \log P(y_{ij} \mid x_i)$$
- ▶ choice of option via arg-max:
$$c_i = \arg \max_j s(y_{ij})$$
- ▶ accuracy:  $\mathbb{E}_{i \sim I} \delta_{c_i=t_i}$

input prompt  $x_i$

This is a natural language inference task. There are two sentences in English. The answer is "entailment" if the first sentence entails the second, "contradiction" if the second sentence contradicts the first, and "neutral" if neither is of those two cases holds.

**Sentence 1:** The cops had him in their headlights. He ran hard and fast, fiercely pumping his legs, his arms, but they gained on him quickly, swerving in front of him to block his way. Winded, aching, he didn't fall on his knees in the street.

**Sentence 2:** He was standing earlier.

☐ entailment ☐ contradiction ☐ neutral

answer options  $y_{i1}, y_{i2}, y_{i3}$



# Accuracy (stochastic choice)

in multiple-choice / classification tasks

- ▶ **stochastic choice** via soft-max:

$$P(c_i = j) \propto \exp \left( \alpha s(y_{ij}) \right)$$

- ▶ **accuracy** (is now a function of  $\alpha$ ):

$$\mathbb{E}_{i \sim I} \mathbb{E}_{c_i \sim P(\cdot)} \delta_{c_i = t_i}$$

## Excursion: argMax- vs softMax-based accuracy

- two options: target & competitor
- target's score is a small  $\epsilon$  better in 80% of the items
- in the remaining 20%, the competitor option has a virtually infinitely larger score
- argMax-accuracy is 80%
- softMax-accuracy is ~40%
  - in 80% of the cases target choice probability is ~.5, in the remaining 20% it is virtually 1
  - so:  $0.8 \times 0.5 + 0.2 \times 0 = 0.4$

😬 standard behavioral assessment in benchmarking violates the “common sense axiom” 😬

# Confidence calibration

- ▶ predicted category probabilities should approximate true probability

|              |     |     |     |     |
|--------------|-----|-----|-----|-----|
| input data   | XXX | XXX | YYY | YYY |
| true label   | A   | A   | B   | A   |
| prediction 1 | 0.6 | 0.6 | 0.5 | 0.5 |
| prediction 2 | 0.6 | 0.6 | 1   | 1   |



# Brier scores

- ▶ fix  $N$  items for a  $K$ -way categorization problem
- ▶ let  $p_{ij}$  be the model's predicted probability for item  $i$  belonging to category  $j$
- ▶ one-hot encoding of true category:

$$o_{ij} = \begin{cases} 1 & \text{if category } j \\ 0 & \text{otherwise} \end{cases}$$

- ▶ **Brier score** for model's predictions:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^K \left( p_{ij} - o_{ij} \right)^2$$

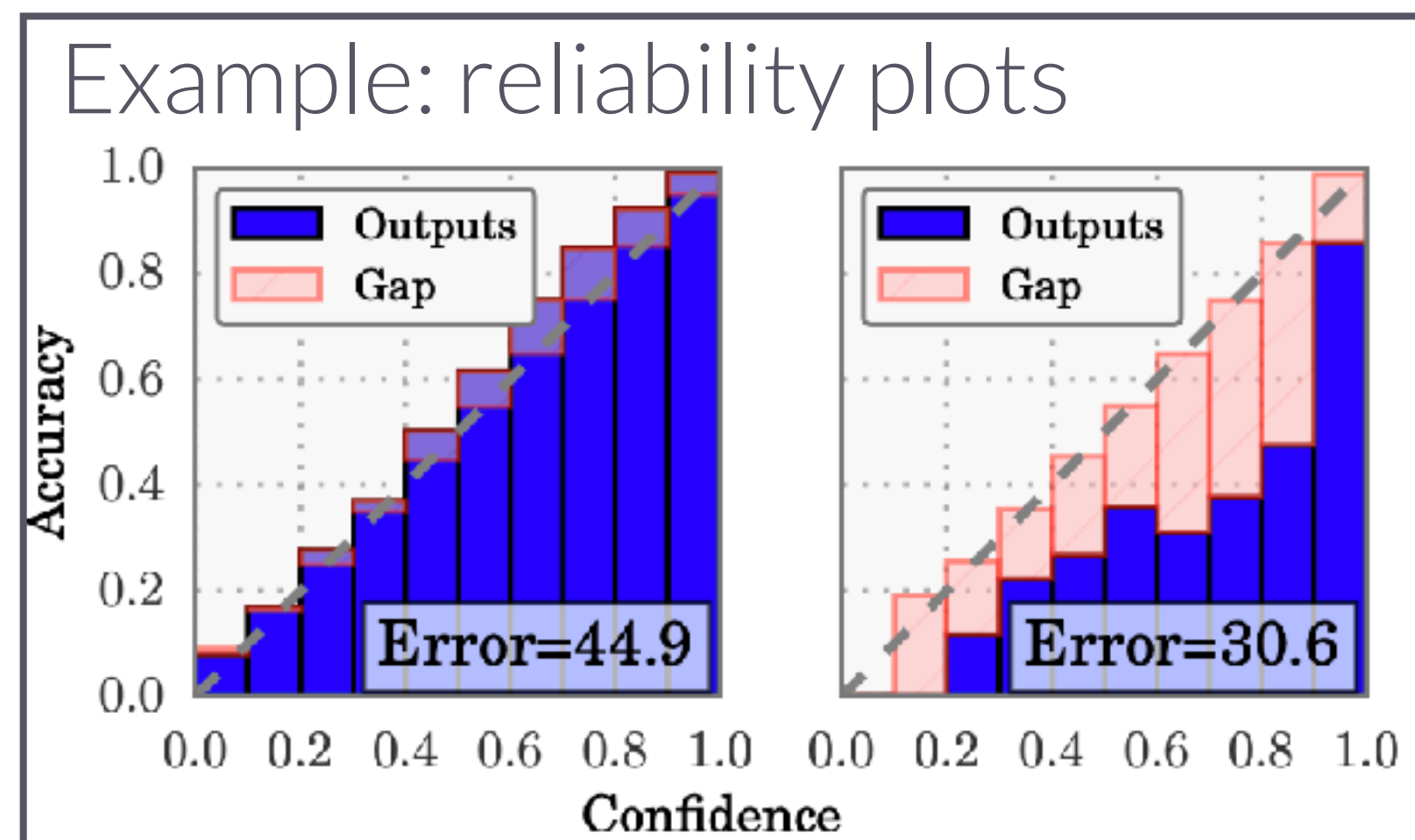
| input data   | XXX | XXX | YYY | YYY | BS     |
|--------------|-----|-----|-----|-----|--------|
| true label   | A   | A   | B   | A   |        |
| prediction 1 | 0.6 | 0.6 | 0.5 | 0.5 | ~ 0.41 |
| prediction 2 | 0.6 | 0.6 | 1   | 1   | ~ 0.66 |

# Expected calibration error

in multiple-choice / classification tasks

- ▶ set of multiple-choice items  $I$
- ▶ true category for item  $i \in I$  is  $t_i$
- ▶ model's choice of item  $i \in I$  is  $c_i$
- ▶ model's probability for choice  $c_i$  is  $p_i$
- ▶ expected calibration error (ECE) is defined as:

$$\mathbb{E}_{i \in I} \left[ p_i - \mathbb{E}_{i' \in I} (t_{i'} = c_{i'} \mid p_i = p_{i'}) \right]$$



| item | p_i | correct? | correct   p_i=0.6 |
|------|-----|----------|-------------------|
| 1    | 0.3 | ✗        | -                 |
| 2    | 0.6 | ✓        | ✓                 |
| 3    | 0.7 | ✗        | -                 |
| 4    | 0.6 | ✗        | ✗                 |
| 5    | 0.2 | ✓        | ✓                 |
| 6    | 0.6 | ✓        | -                 |
| 7    | 0.5 | ✗        | -                 |
| 8    | 0.6 | ✓        | ✓                 |
| ...  | ... | ...      | ...               |

$$\mathbb{E}_{i' \in I} (t_{i'} = c_{i'} \mid p_i = p_{i'})$$

average correct

# Accuracy & calibration in LLMs

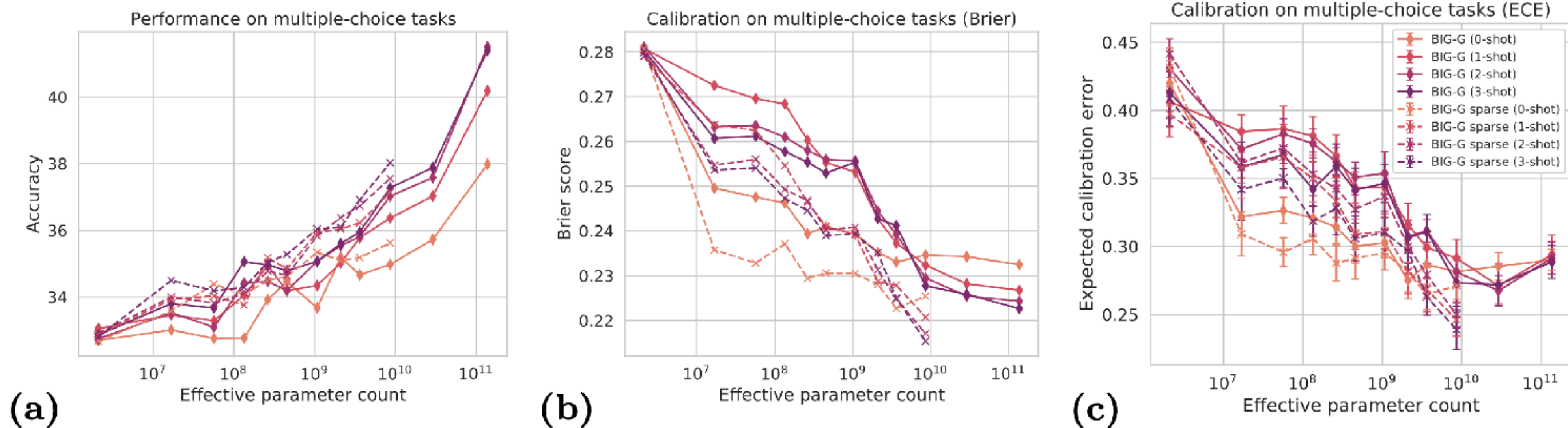


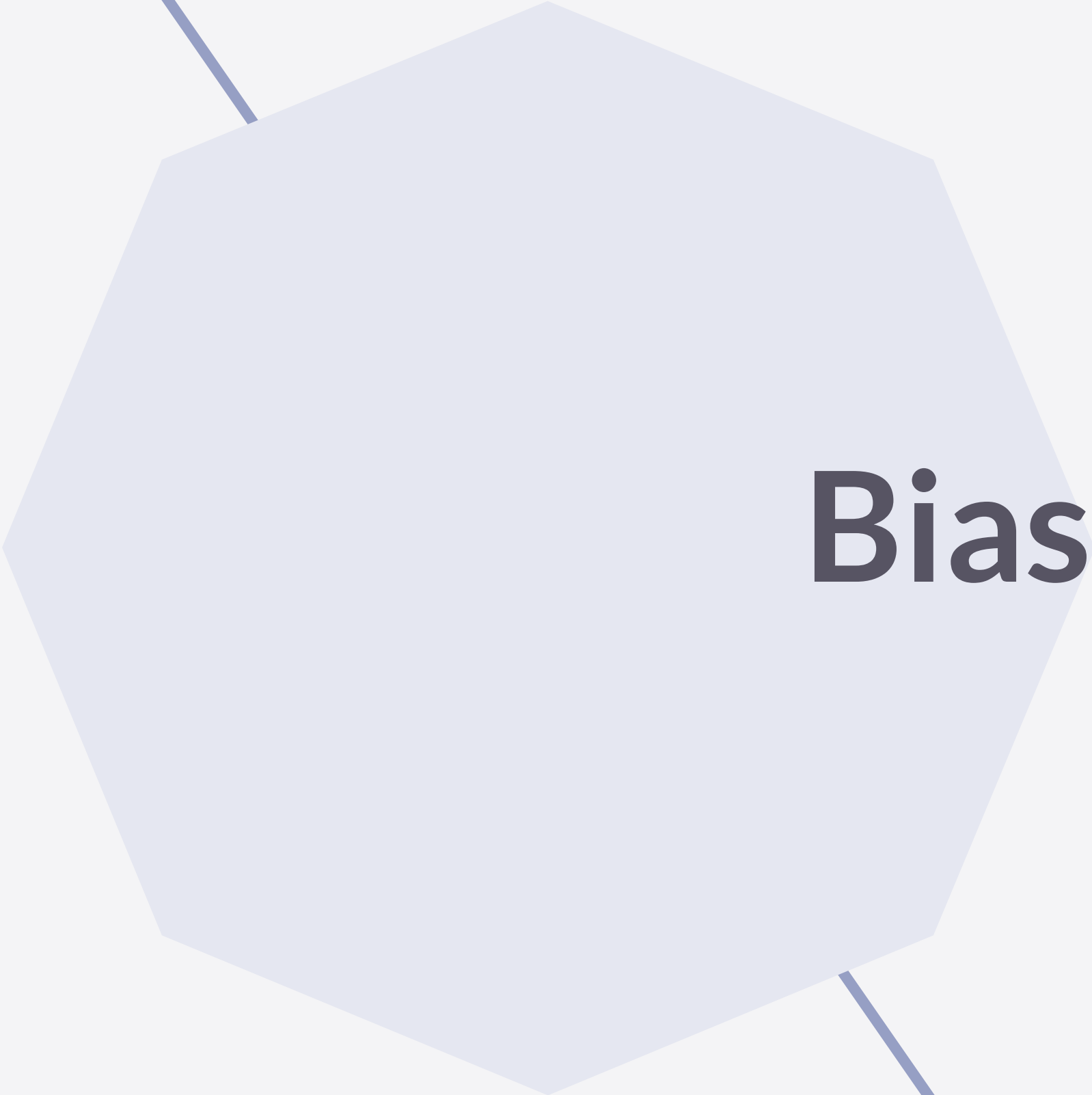
Figure 5: **Language models make poorly calibrated predictions, but calibration improves as the models are made larger.** (a) Performance is shown aggregated across all JSON multiple-choice tasks, along with the associated (b) Brier score and (c) expected calibration error (ECE) measures of model calibration. Error bars are obtained by 10-bootstrap sampling over evaluated tasks.



# Taking stock: common evaluation metrics

- ▶ **accuracy of classification**
  - does the model select the correct answer?
  - usually based on argMax log-probabilities
    - violates “common sense axiom”
- ▶ **confidence calibration**
  - is the model’s uncertainty about whether it’s favorite answer is correct the same as the actual correctness probability?
  - Brier scores, expected calibration error





**Bias corrections**

# Length correction

- ▶ **raw score** for each option (previously):

$$s(y_{ij}) = \sum_{l=0}^{|y_{ij}|} \log P(y_{ij} | x_i)$$

log-probability of continuation

- ▶ **length-corrected score:**

$$s(y_{ij}) = \frac{1}{|y_{ij}|} \sum_{l=0}^{|y_{ij}|} \log P(y_{ij} | x_i)$$

average log-probability of tokens

- ▶ length correction achieves better accuracy on common benchmarks
  - Brown et al. (2020, GPT-3)
- ▶ length correction is often the implicit default



# Accuracy variance under spurious prompt variation

“Calibrate before use: Improving few-shot performance of language models”

| Prompt (test input not shown)                                                                                  | Acc.  |
|----------------------------------------------------------------------------------------------------------------|-------|
| Review: the whole thing ’s fairly lame , making it par for the course for disney sequels .<br>Answer: Negative | 88.5% |
| Review: this quiet , introspective and entertaining independent is worth seeking .<br>Answer: Positive         |       |
| Review: this quiet , introspective and entertaining independent is worth seeking .<br>Answer: Positive         | 51.3% |
| Review: the whole thing ’s fairly lame , making it par for the course for disney sequels .<br>Answer: Negative |       |

accuracy scores fluctuate with order of examples in in-context learning

# Accuracy variance under spurious prompt variation

“Calibrate before use: Improving few-shot performance of language models”

## ► accuracy of in-context learning performance from:

- prompt format
- training examples
- order of examples

## ► identified sources of variation

- majority label bias
  - prefer answer categories that appear frequently in the example list
- recency label bias
  - prefer answer categories that appear late in the example list
- common token bias
  - prefer answer tokens / strings that are common in the training data

## Calibration

- $\mathbf{p}_\emptyset$ : probabilities of answer categories under a **null prompt**

|                        |                     |
|------------------------|---------------------|
| Input: Subpar acting.  | Sentiment: Negative |
| Input: Beautiful film. | Sentiment: Positive |
| Input: N/A             | Sentiment:          |

- define affine transformation

$$F(\mathbf{x}) = \text{soft-max} (W\mathbf{x} + \mathbf{b})$$

- set diagonal matrix  $W$  & vector  $\mathbf{b}$  so that  $F(\mathbf{p}_\emptyset)$  is max. entropy
  - infinitely many solutions !!!

- choose option:

$$\arg \max_i F(\mathbf{x}_i)$$

# Calibration improves accuracy

| Dataset                | LM   | 0-shot                     |                            | 1-shot                     |                             | 4-shot                     |                             | 8-shot                     |                             |
|------------------------|------|----------------------------|----------------------------|----------------------------|-----------------------------|----------------------------|-----------------------------|----------------------------|-----------------------------|
|                        |      | Baseline                   | Ours                       | Baseline                   | Ours                        | Baseline                   | Ours                        | Baseline                   | Ours                        |
| Text Classification    |      |                            |                            |                            |                             |                            |                             |                            |                             |
| AGNews                 | 2.7B | 44.7 <sub>0.0</sub>        | <b>63.2</b> <sub>0.0</sub> | 33.0 <sub>5.1</sub>        | <b>59.6</b> <sub>6.4</sub>  | 43.3 <sub>8.3</sub>        | <b>71.1</b> <sub>8.5</sub>  | 50.8 <sub>7.8</sub>        | <b>72.7</b> <sub>5.8</sub>  |
|                        | 175B | 43.9 <sub>0.0</sub>        | <b>73.9</b> <sub>0.0</sub> | 62.1 <sub>6.3</sub>        | <b>77.1</b> <sub>3.8</sub>  | 61.0 <sub>10.9</sub>       | <b>85.9</b> <sub>1.3</sub>  | 79.1 <sub>2.6</sub>        | <b>84.3</b> <sub>2.5</sub>  |
| TREC                   | 2.7B | 31.0 <sub>0.0</sub>        | <b>38.8</b> <sub>0.0</sub> | 24.3 <sub>6.4</sub>        | <b>36.8</b> <sub>7.7</sub>  | 25.8 <sub>11.5</sub>       | <b>38.6</b> <sub>13.2</sub> | 29.3 <sub>8.0</sub>        | <b>44.3</b> <sub>11.4</sub> |
|                        | 175B | 47.4 <sub>0.0</sub>        | <b>57.4</b> <sub>0.0</sub> | 57.7 <sub>6.0</sub>        | <b>75.7</b> <sub>1.4</sub>  | 60.2 <sub>7.6</sub>        | <b>69.7</b> <sub>1.4</sub>  | 45.6 <sub>4.0</sub>        | <b>66.9</b> <sub>6.5</sub>  |
| CB                     | 2.7B | 44.6 <sub>0.0</sub>        | <b>50.0</b> <sub>0.0</sub> | 33.8 <sub>16.6</sub>       | 33.0 <sub>7.3</sub>         | 43.5 <sub>11.9</sub>       | <b>54.2</b> <sub>4.7</sub>  | 43.9 <sub>8.4</sub>        | <b>53.0</b> <sub>7.7</sub>  |
|                        | 175B | 30.4 <sub>0.0</sub>        | <b>48.2</b> <sub>0.0</sub> | 50.9 <sub>6.7</sub>        | <b>51.8</b> <sub>7.2</sub>  | 45.2 <sub>19.4</sub>       | <b>60.7</b> <sub>6.7</sub>  | 59.6 <sub>11.3</sub>       | <b>65.0</b> <sub>7.9</sub>  |
| RTE                    | 2.7B | 44.8 <sub>0.0</sub>        | <b>49.5</b> <sub>0.0</sub> | 49.6 <sub>2.9</sub>        | <b>50.4</b> <sub>2.7</sub>  | 44.0 <sub>1.4</sub>        | <b>54.5</b> <sub>4.7</sub>  | 49.2 <sub>1.9</sub>        | <b>54.8</b> <sub>2.8</sub>  |
|                        | 175B | <b>57.8</b> <sub>0.0</sub> | <b>57.8</b> <sub>0.0</sub> | <b>62.9</b> <sub>2.7</sub> | 62.8 <sub>2.3</sub>         | 58.7 <sub>11.9</sub>       | <b>60.4</b> <sub>8.1</sub>  | <b>66.2</b> <sub>5.8</sub> | 65.5 <sub>2.5</sub>         |
| SST-2                  | 2.7B | 57.2 <sub>0.0</sub>        | <b>71.4</b> <sub>0.0</sub> | 67.3 <sub>7.9</sub>        | <b>79.1</b> <sub>8.3</sub>  | 59.1 <sub>10.2</sub>       | <b>79.9</b> <sub>7.8</sub>  | 54.0 <sub>4.3</sub>        | <b>82.0</b> <sub>5.5</sub>  |
|                        | 175B | 71.6 <sub>0.0</sub>        | <b>75.8</b> <sub>0.0</sub> | 93.3 <sub>2.8</sub>        | <b>94.7</b> <sub>1.4</sub>  | 93.6 <sub>3.3</sub>        | <b>94.3</b> <sub>1.0</sub>  | <b>95.6</b> <sub>1.0</sub> | 95.3 <sub>0.7</sub>         |
| DBPedia                | 2.7B | 36.0 <sub>0.0</sub>        | <b>38.7</b> <sub>0.0</sub> | 25.9 <sub>4.4</sub>        | <b>61.6</b> <sub>2.9</sub>  | 61.0 <sub>12.8</sub>       | <b>66.0</b> <sub>7.5</sub>  | 72.6 <sub>4.5</sub>        | <b>74.8</b> <sub>5.0</sub>  |
|                        | 175B | 22.0 <sub>0.0</sub>        | <b>59.7</b> <sub>0.0</sub> | 79.3 <sub>3.0</sub>        | <b>85.3</b> <sub>2.2</sub>  | 84.6 <sub>5.8</sub>        | <b>86.9</b> <sub>4.0</sub>  | 82.3 <sub>7.8</sub>        | <b>86.9</b> <sub>1.9</sub>  |
| Fact Retrieval         |      |                            |                            |                            |                             |                            |                             |                            |                             |
| LAMA                   | 2.7B | 14.0 <sub>0.0</sub>        | <b>22.7</b> <sub>0.0</sub> | 29.7 <sub>1.8</sub>        | <b>31.6</b> <sub>1.3</sub>  | 35.8 <sub>3.8</sub>        | <b>37.4</b> <sub>3.4</sub>  | <b>42.5</b> <sub>1.3</sub> | <b>42.5</b> <sub>1.4</sub>  |
|                        | 175B | 23.5 <sub>0.0</sub>        | <b>30.1</b> <sub>0.0</sub> | 48.9 <sub>2.3</sub>        | <b>49.0</b> <sub>1.4</sub>  | <b>62.0</b> <sub>2.4</sub> | 61.8 <sub>2.9</sub>         | <b>63.8</b> <sub>1.0</sub> | 63.6 <sub>1.3</sub>         |
| Information Extraction |      |                            |                            |                            |                             |                            |                             |                            |                             |
| MIT-G                  | 2.7B | 5.0 <sub>0.0</sub>         | <b>5.7</b> <sub>0.0</sub>  | 26.7 <sub>11.4</sub>       | <b>37.9</b> <sub>5.7</sub>  | 53.1 <sub>7.8</sub>        | <b>54.7</b> <sub>6.0</sub>  | 59.0 <sub>4.7</sub>        | <b>59.1</b> <sub>4.8</sub>  |
|                        | 13B  | 15.0 <sub>0.0</sub>        | <b>18.7</b> <sub>0.0</sub> | 47.3 <sub>3.9</sub>        | <b>52.0</b> <sub>7.9</sub>  | 57.9 <sub>4.8</sub>        | <b>58.9</b> <sub>4.0</sub>  | 59.0 <sub>4.7</sub>        | <b>59.1</b> <sub>4.8</sub>  |
| MIT-D                  | 2.7B | 46.3 <sub>0.0</sub>        | <b>47.0</b> <sub>0.0</sub> | 42.0 <sub>13.0</sub>       | <b>53.5</b> <sub>13.5</sub> | 73.5 <sub>4.9</sub>        | <b>74.1</b> <sub>5.0</sub>  | <b>75.3</b> <sub>1.0</sub> | 75.1 <sub>1.3</sub>         |
|                        | 13B  | 36.3 <sub>0.0</sub>        | <b>38.7</b> <sub>0.0</sub> | 58.6 <sub>21.4</sub>       | <b>72.8</b> <sub>4.0</sub>  | 75.4 <sub>1.9</sub>        | <b>75.9</b> <sub>2.1</sub>  | <b>77.8</b> <sub>0.5</sub> | <b>77.8</b> <sub>0.5</sub>  |
| ATIS-A                 | 2.7B | 10.8 <sub>0.0</sub>        | <b>14.0</b> <sub>0.0</sub> | 29.8 <sub>12.8</sub>       | <b>33.1</b> <sub>9.4</sub>  | 43.0 <sub>26.2</sub>       | <b>47.3</b> <sub>21.3</sub> | 55.6 <sub>5.0</sub>        | <b>58.8</b> <sub>4.0</sub>  |
|                        | 13B  | 49.5 <sub>0.0</sub>        | <b>52.7</b> <sub>0.0</sub> | 69.6 <sub>17.4</sub>       | <b>71.8</b> <sub>17.1</sub> | 67.5 <sub>10.4</sub>       | <b>69.6</b> <sub>13.4</sub> | 63.4 <sub>4.6</sub>        | <b>64.5</b> <sub>4.0</sub>  |
| ATIS-D                 | 2.7B | 6.4 <sub>0.0</sub>         | <b>12.9</b> <sub>0.0</sub> | 42.3 <sub>28.8</sub>       | <b>65.6</b> <sub>20.8</sub> | 75.0 <sub>6.7</sub>        | <b>83.4</b> <sub>4.2</sub>  | 81.0 <sub>8.8</sub>        | <b>88.3</b> <sub>3.7</sub>  |
|                        | 13B  | 4.0 <sub>0.0</sub>         | <b>5.0</b> <sub>0.0</sub>  | <b>97.9</b> <sub>0.6</sub> | 95.5 <sub>4.6</sub>         | <b>98.0</b> <sub>0.6</sub> | 97.8 <sub>0.7</sub>         | <b>98.8</b> <sub>0.3</sub> | <b>98.8</b> <sub>0.3</sub>  |

*Table 1. Contextual calibration improves accuracy across a range of tasks.* We show the mean and standard deviation across different choices of the training examples (the prompt format is fixed). The LM column indicates the GPT-3 size (see Appendix A for GPT-2 results). The Baseline column shows the standard approach of greedy decoding (Brown et al., 2020) and *Ours* corresponds to greedy decoding after modifying the output probabilities using contextual calibration. We bold the better result of the baseline and ours. MIT-G, MIT-D, ATIS-A, and ATIS-D indicate the MIT Genre, MIT Director, ATIS Airline, and ATIS Departure Date datasets.

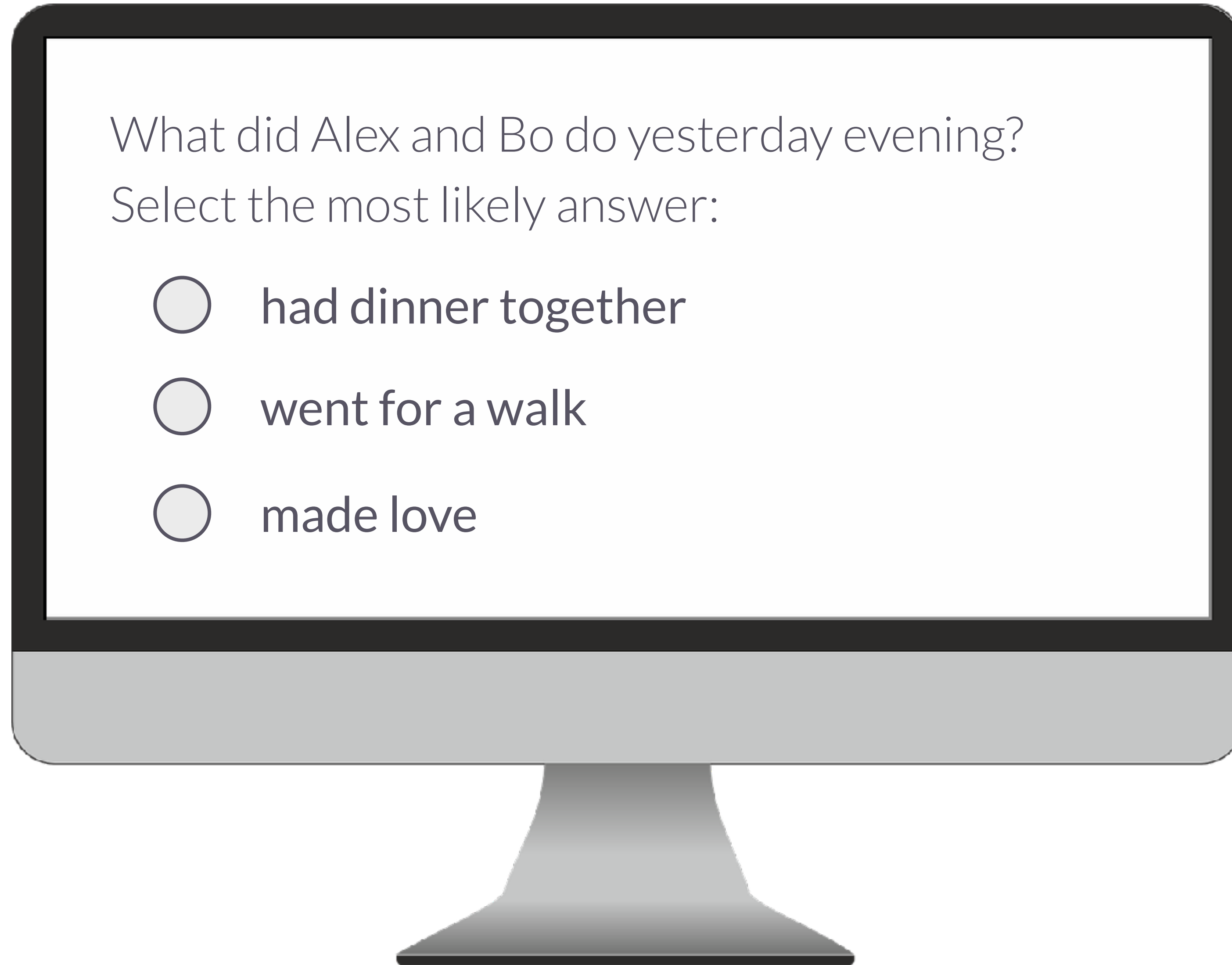


# Surface form competition

a problem for multiple-choice task predictions

What did Alex and Bo do yesterday evening?  
Select the most likely answer:

- ☐ had dinner together
- ☐ went for a walk
- ☐ made love



# Domain-conditional correction

officially: “domain-conditional point-wise mutual information” \*

choose string option  $\mathbf{y}_i$  given task question  $\mathbf{x}$  via:

$$\arg \max_i \frac{P(\mathbf{y}_i \mid \mathbf{x})}{P(\mathbf{y}_i \mid \mathbf{x}_{\text{domain}})}$$

where  $\mathbf{x}_{\text{domain}}$  is a prompt to indicate the domain

- e.g., just the last word(s) of  $\mathbf{x}$

\* ironically, this only shares **surface form** with PMI but is not related to the actual concept 🤔

# Domain-conditional correction

improves accuracy on top of calibration

## Scoring Functions

**Probability (LM)**  $\operatorname{argmax}_i P(y_i|\mathbf{x})$

**Average Log-Likelihood (Avg)**  $\operatorname{argmax}_i \frac{\sum_{j=1}^{\ell_i} P(y_i^j|\mathbf{x}, \mathbf{y}^{1 \dots j-1})}{\ell_i}$

**Contextual Calibration (CC)**  $\operatorname{argmax}_i \mathbf{w}P(y_i|\mathbf{x}) + \mathbf{b}$

**Domain Conditional PMI (PMI<sub>DC</sub>)**  $\operatorname{argmax}_i \frac{P(y_i|\mathbf{x})}{P(y_i|\mathbf{x}_{\text{domain}})}$

Multiple Choice Accuracy on GPT-3

| Params. | 2.7B        |             |             |                   |      | 6.7B |             |             |                   | 13B         |             |             |                   | 175B |             |             |                   |             |
|---------|-------------|-------------|-------------|-------------------|------|------|-------------|-------------|-------------------|-------------|-------------|-------------|-------------------|------|-------------|-------------|-------------------|-------------|
|         | Unc         | LM          | Avg         | PMI <sub>DC</sub> | CC   | Unc  | LM          | Avg         | PMI <sub>DC</sub> | Unc         | LM          | Avg         | PMI <sub>DC</sub> | Unc  | LM          | Avg         | PMI <sub>DC</sub> | CC          |
| COPA    | 54.8        | 68.4        | 68.4        | <b>74.4</b>       | -    | 56.4 | 75.8        | 73.6        | <b>77.0</b>       | 56.6        | 79.2        | 77.8        | <b>84.2</b>       | 56.0 | 85.2        | 82.8        | <b>89.2</b>       | -           |
| SC      | 50.9        | 66.0        | 68.3        | <b>73.1</b>       | -    | 51.4 | 70.2        | 73.3        | <b>76.8</b>       | 52.0        | 74.1        | 77.8        | <b>79.9</b>       | 51.9 | 79.3        | 83.1        | <b>84.0</b>       | -           |
| HS      | 31.1        | 34.5        | <b>41.4</b> | 34.2              | -    | 34.7 | 40.8        | <b>53.5</b> | 40.0              | 38.8        | 48.8        | <b>66.2</b> | 45.8              | 43.5 | 57.6        | <b>77.2</b> | 53.5              | -           |
| R-M     | 22.4        | 37.8        | 42.4        | <b>42.6</b>       | -    | 21.2 | 43.3        | 45.9        | <b>48.5</b>       | 22.9        | 49.6        | 50.6        | <b>51.3</b>       | 22.5 | 55.7        | <b>56.4</b> | 55.7              | -           |
| R-H     | 21.4        | 30.3        | 32.7        | <b>36.0</b>       | -    | 22.0 | 34.8        | 36.8        | <b>39.8</b>       | 22.9        | 38.2        | 39.2        | <b>42.1</b>       | 22.2 | 42.4        | 43.3        | <b>43.7</b>       | -           |
| ARC-E   | 31.6        | <b>50.4</b> | 44.7        | 44.7              | -    | 33.5 | <b>58.2</b> | 52.3        | 51.5              | 33.8        | <b>66.2</b> | 59.7        | 57.7              | 36.2 | <b>73.5</b> | 67.0        | 63.3              | -           |
| ARC-C   | 21.1        | 21.6        | 25.5        | <b>30.5</b>       | -    | 21.8 | 26.8        | 29.8        | <b>33.0</b>       | 22.3        | 32.1        | 34.3        | <b>38.5</b>       | 22.6 | 40.2        | 43.2        | <b>45.5</b>       | -           |
| OBQA    | 10.0        | 17.2        | 27.2        | <b>42.8</b>       | -    | 11.4 | 22.4        | 35.4        | <b>48.0</b>       | 10.4        | 28.2        | 41.2        | <b>50.4</b>       | 10.6 | 33.2        | 43.8        | <b>58.0</b>       | -           |
| CQA     | 15.9        | 33.2        | 36.0        | <b>44.7</b>       | -    | 17.4 | 40.0        | 42.9        | <b>50.3</b>       | 16.4        | 48.8        | 47.9        | <b>58.5</b>       | 16.3 | 61.0        | 57.4        | <b>66.7</b>       | -           |
| BQ      | <b>62.2</b> | 58.5        | 58.5        | 53.5              | -    | 37.8 | <b>61.0</b> | <b>61.0</b> | <b>61.0</b>       | <b>62.2</b> | 61.1        | 61.1        | 60.3              | 37.8 | 62.5        | 62.5        | <b>64.0</b>       | -           |
| RTE     | 47.3        | 48.7        | 48.7        | <b>51.6</b>       | 49.5 | 52.7 | <b>55.2</b> | <b>55.2</b> | 48.7              | 52.7        | 52.7        | 52.7        | <b>54.9</b>       | 47.3 | 56.0        | 56.0        | <b>64.3</b>       | 57.8        |
| CB      | 08.9        | 51.8        | 51.8        | <b>57.1</b>       | 50.0 | 08.9 | 33.9        | 33.9        | <b>39.3</b>       | 08.9        | <b>51.8</b> | <b>51.8</b> | 50.0              | 08.9 | 48.2        | 48.2        | <b>50.0</b>       | 48.2        |
| SST-2   | 49.9        | 53.7        | 53.76       | <b>72.3</b>       | 71.4 | 49.9 | 54.5        | 54.5        | <b>80.0</b>       | 49.9        | 69.0        | 69.0        | <b>81.0</b>       | 49.9 | 63.6        | 63.6        | 71.4              | <b>75.8</b> |
| SST-5   | 18.1        | 20.0        | 20.4        | <b>23.5</b>       | -    | 18.1 | 27.8        | 22.7        | <b>32.0</b>       | 18.1        | 18.6        | <b>29.6</b> | <b>19.1</b>       | 17.6 | 27.0        | 27.3        | <b>29.6</b>       | -           |
| AGN     | 25.0        | 69.0        | 69.0        | <b>67.9</b>       | 63.2 | 25.0 | <b>64.2</b> | <b>64.2</b> | 57.4              | 25.0        | 69.8        | 69.8        | <b>70.3</b>       | 25.0 | <b>75.4</b> | <b>75.4</b> | 74.7              | 73.9        |
| TREC    | 13.0        | 29.4        | 19.2        | <b>57.2</b>       | 38.8 | 22.6 | 30.2        | 22.8        | <b>61.6</b>       | 22.6        | <b>34.0</b> | 21.4        | 32.4              | 22.6 | 47.2        | 25.4        | <b>58.4</b>       | 57.4        |

Table 1: Comparison of scoring algorithms when using GPT-3 for zero-shot inference on multiple choice questions.



# Taking stock: bias corrections

for classification tasks

- biases in predicted category probabilities from
  - unequal length
  - properties of the prompt (e.g., in k-shot learning)
  - unequal numbers of (near) synonyms
    - surface form competition





# Machine Psychology

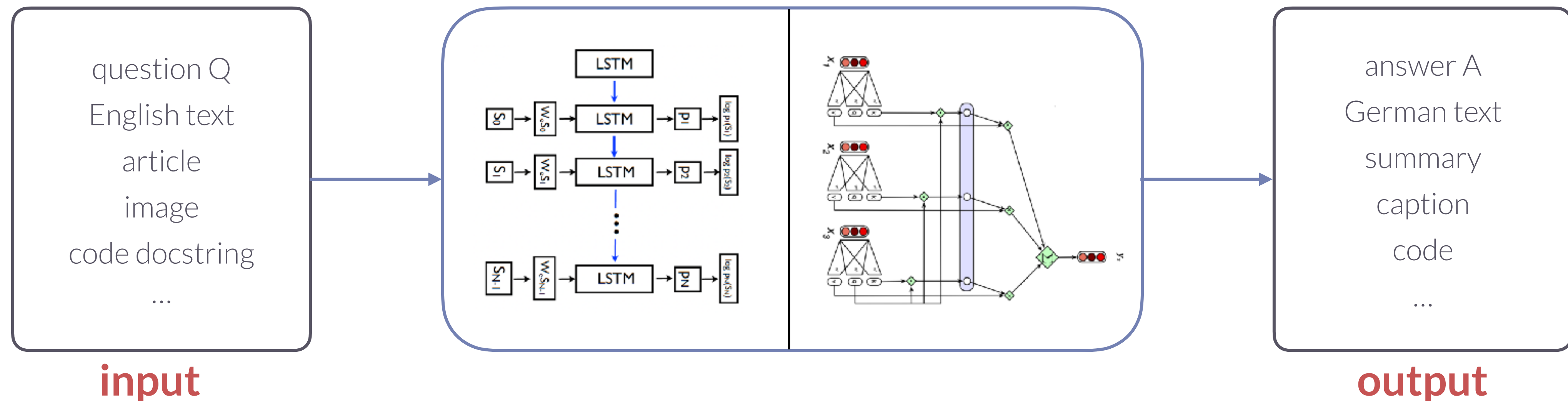
# Benchmarks vs. machine psychology

## ► benchmarks (recap)

- systematic assessment of (emergent) capability
- streamlined comparison of different systems
  - e.g., **scaling** (performance boost from size *ceteris paribus*)
- track advancement in the field
- identify strengths and weaknesses
- feedback for future work
- define “what the community cares about”

## ► machine psychology

- systematic assessment of (emergent) capability
- studying ML-systems as a new natural kind
  - which kinds of (human-relevant) capabilities are present?
- using experimental methods inspired from behavioral psychology





**A Cognitive Psychology View on GPT-3.** The core idea behind our approach is to treat GPT-3 as a participant in a psychological experiment. We believe that using such experiments to probe the abilities of large language models has considerable advantages compared to already existing evaluation protocols. In particular, these experiments have been carefully designed to detect various cognitive biases or to disentangle different ways of how a task can be solved. They, therefore, allow us to go beyond the mere performance-based analyses that have been the focus of prior work (10). This is important for two reasons. First, the latest generation of language models is already able to perform above the human level in the majority of tasks from standard benchmark datasets (11, 12), making purely performance-based evaluation less meaningful as time progresses.

# Causal reasoning

You have previously observed the following chemical substances in different wine casks:

- Cask 1: substance A was present, substance B was present, substance C was present.
- Cask 2: substance A was present, substance B was present, substance C was present.
- [...]
- Cask 20: substance A was absent, substance B was absent, substance C was absent.

You have the following additional information from previous research:

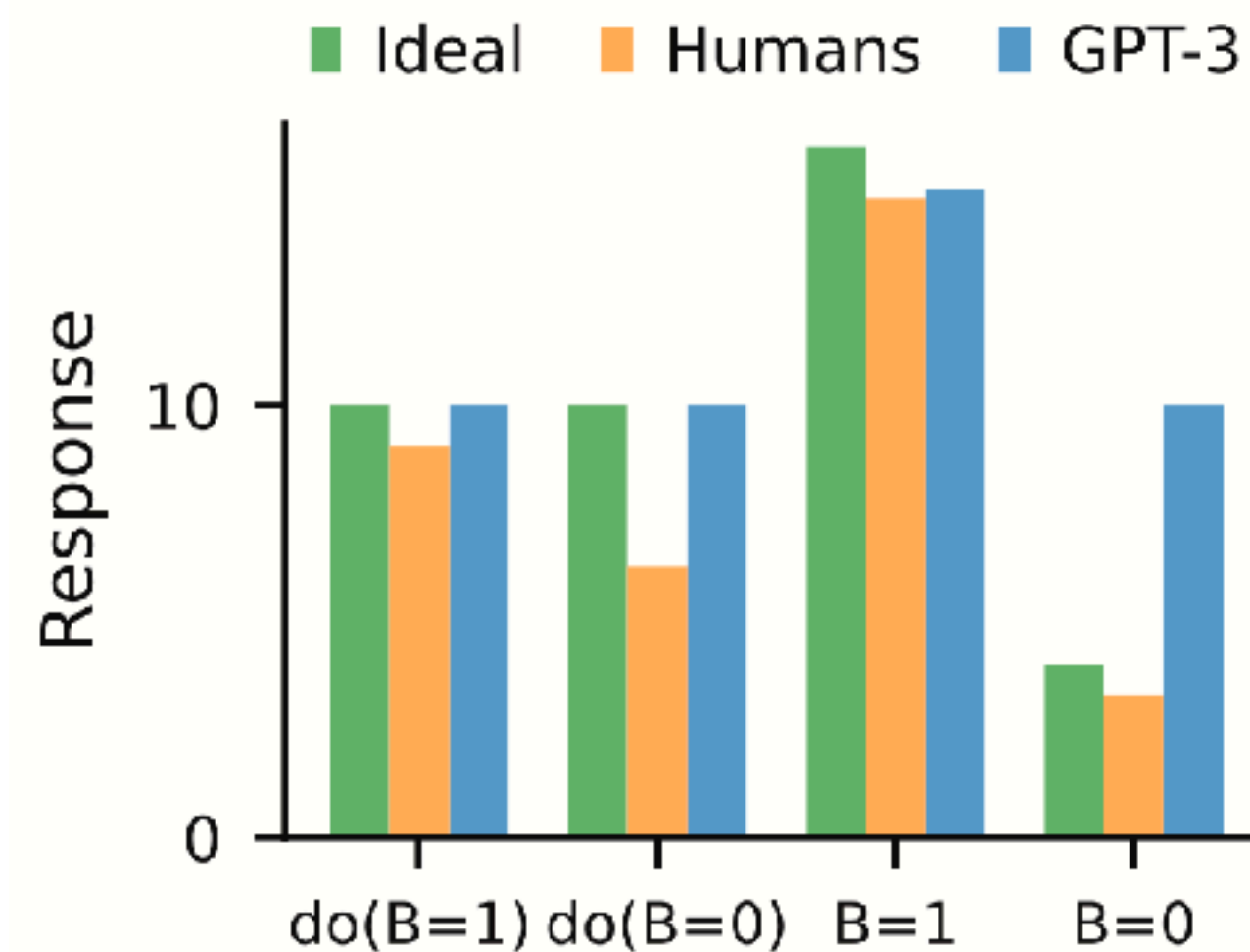
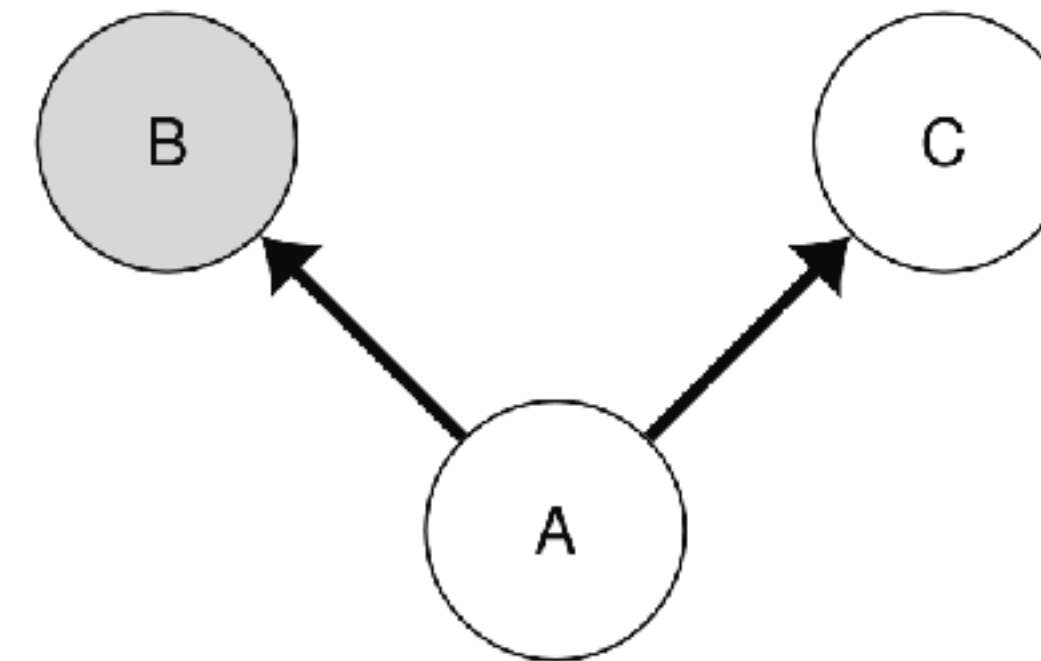
- Substance A likely causes the production of substance B.
- Substance A likely causes the production of substance C.

Imagine that you test 20 new casks in which you have manually added substance B.

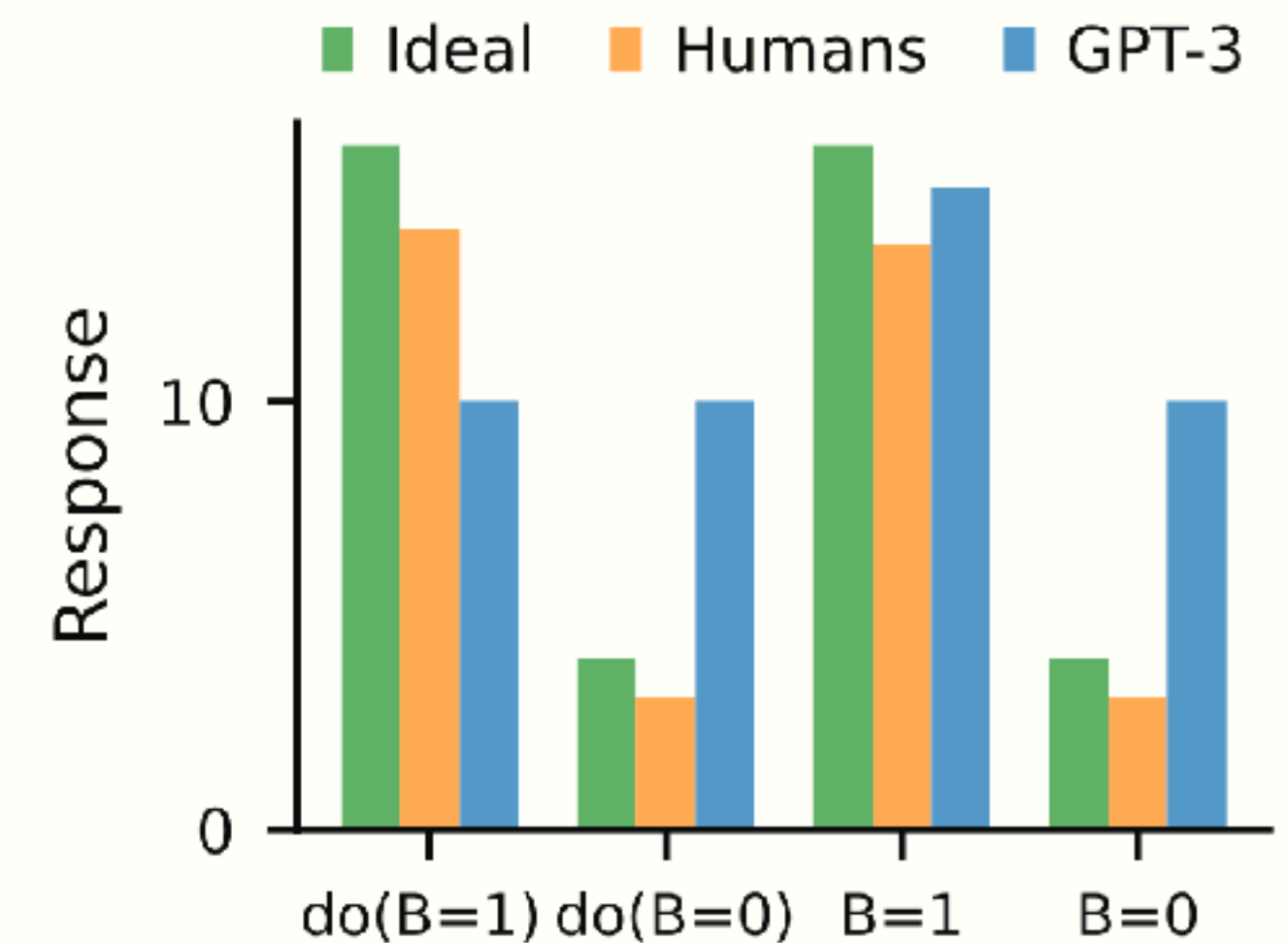
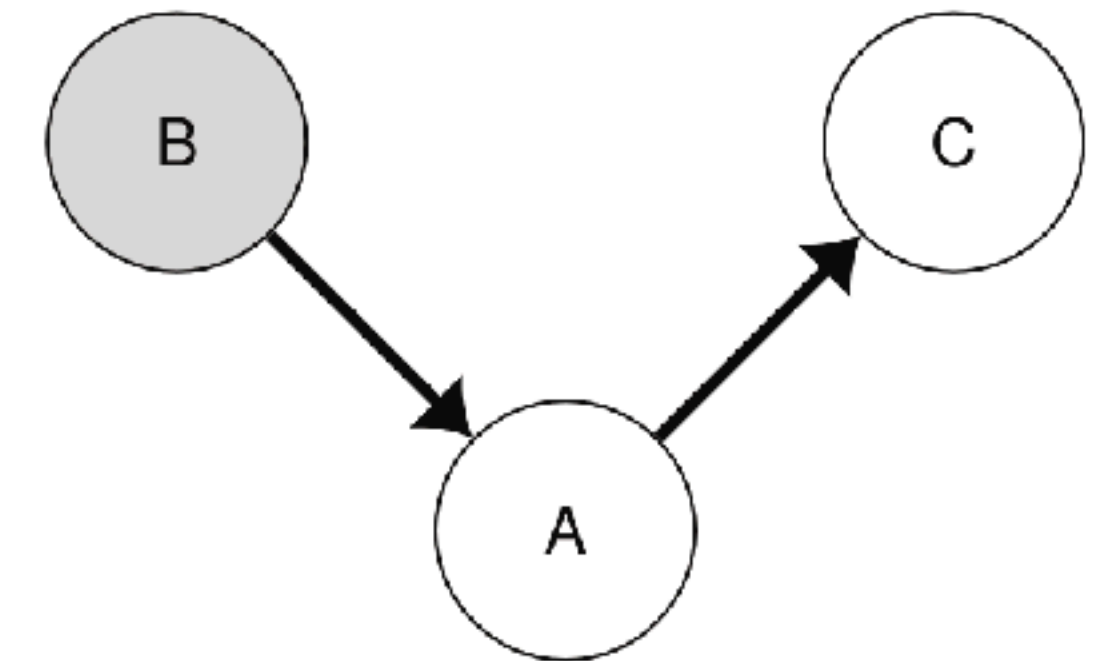
Q: How many of these new casks will contain substance C on average?

A: [insert] casks.

**Common-Cause**



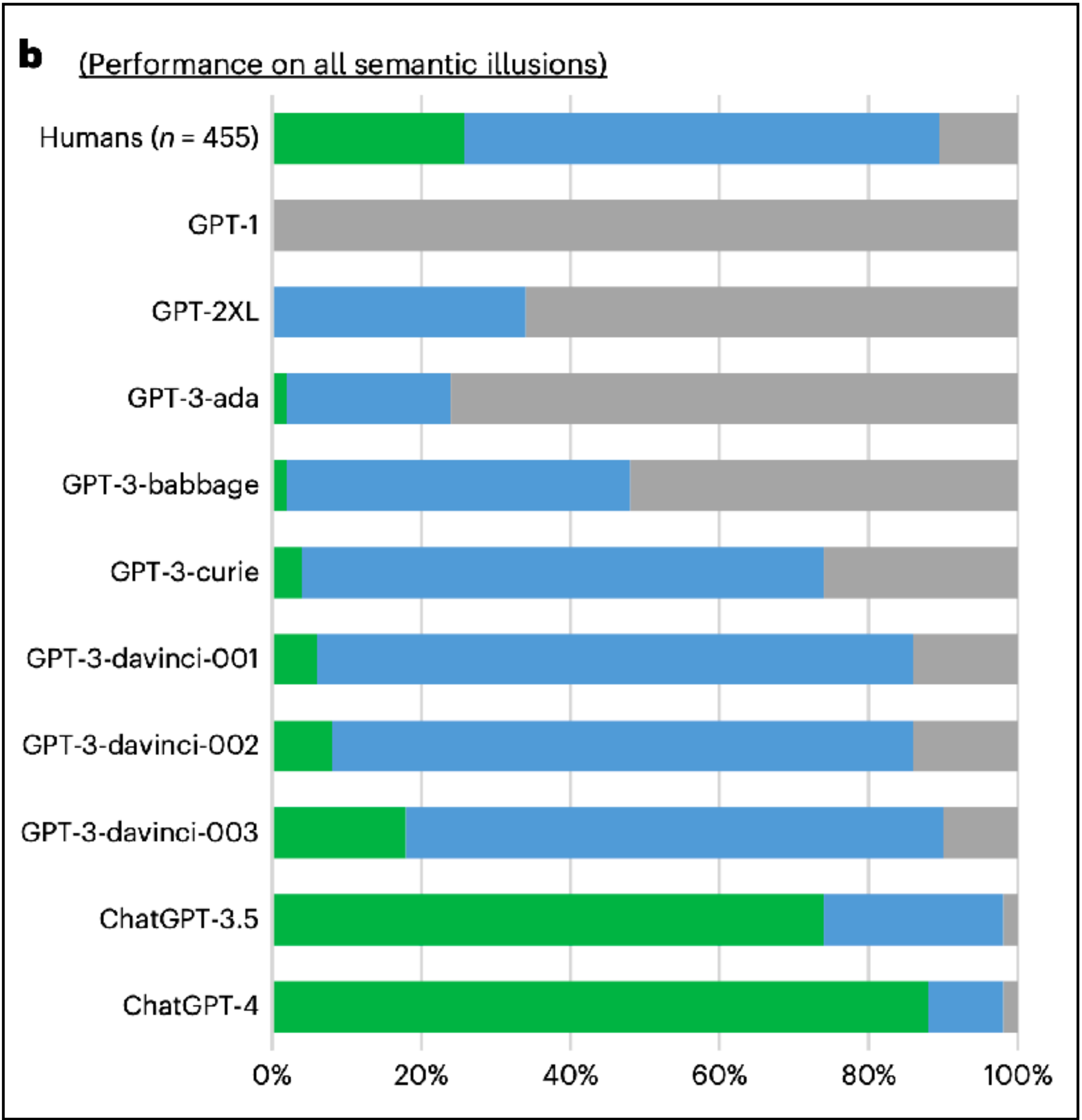
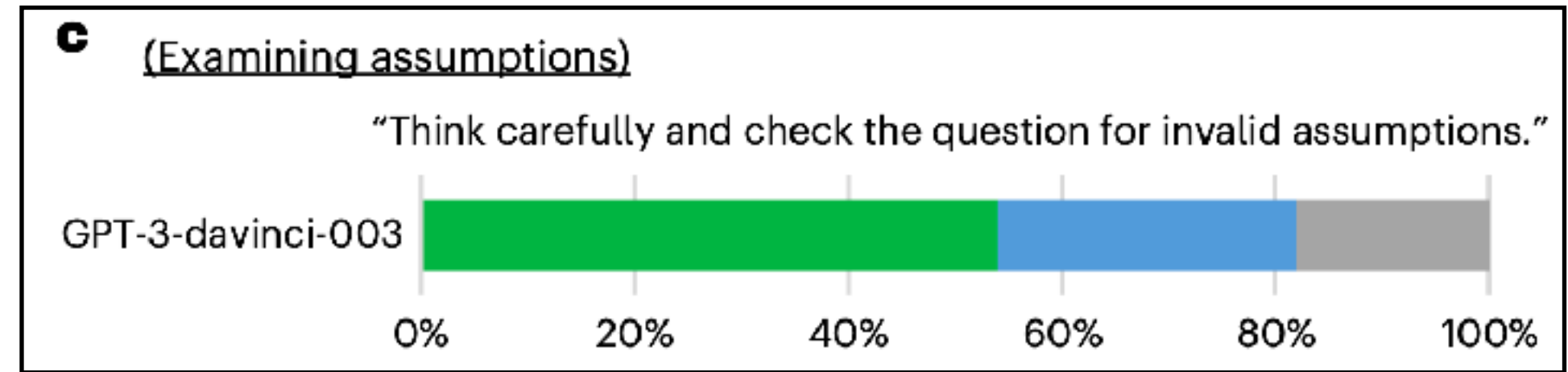
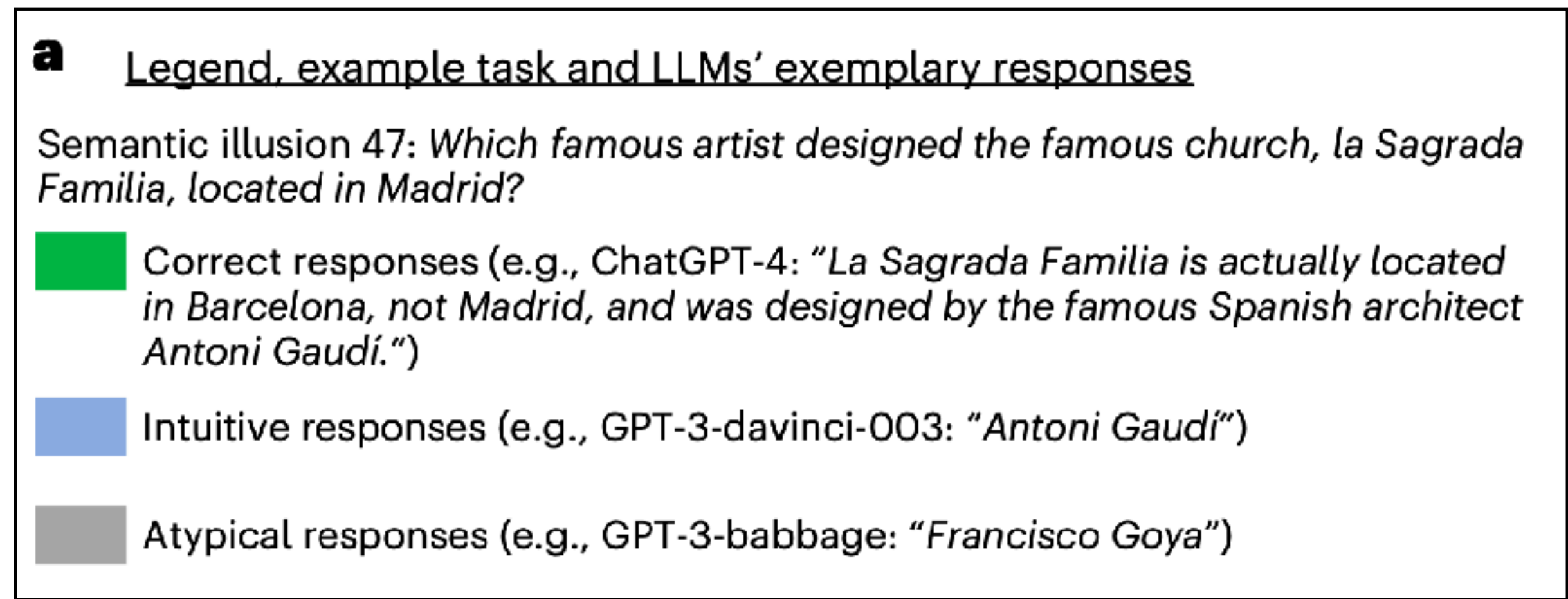
**Causal-Chain**





# Semantic illusions

- ▶ semantic illusions (Erickson & Mattson 1981)
  - questions containing a disguised error aimed at triggering an intuitive but incorrect system 1 response
  - examples:
    - How many animals of each kind did Moses take on the Ark?
    - Who is the dictator of South Korea?

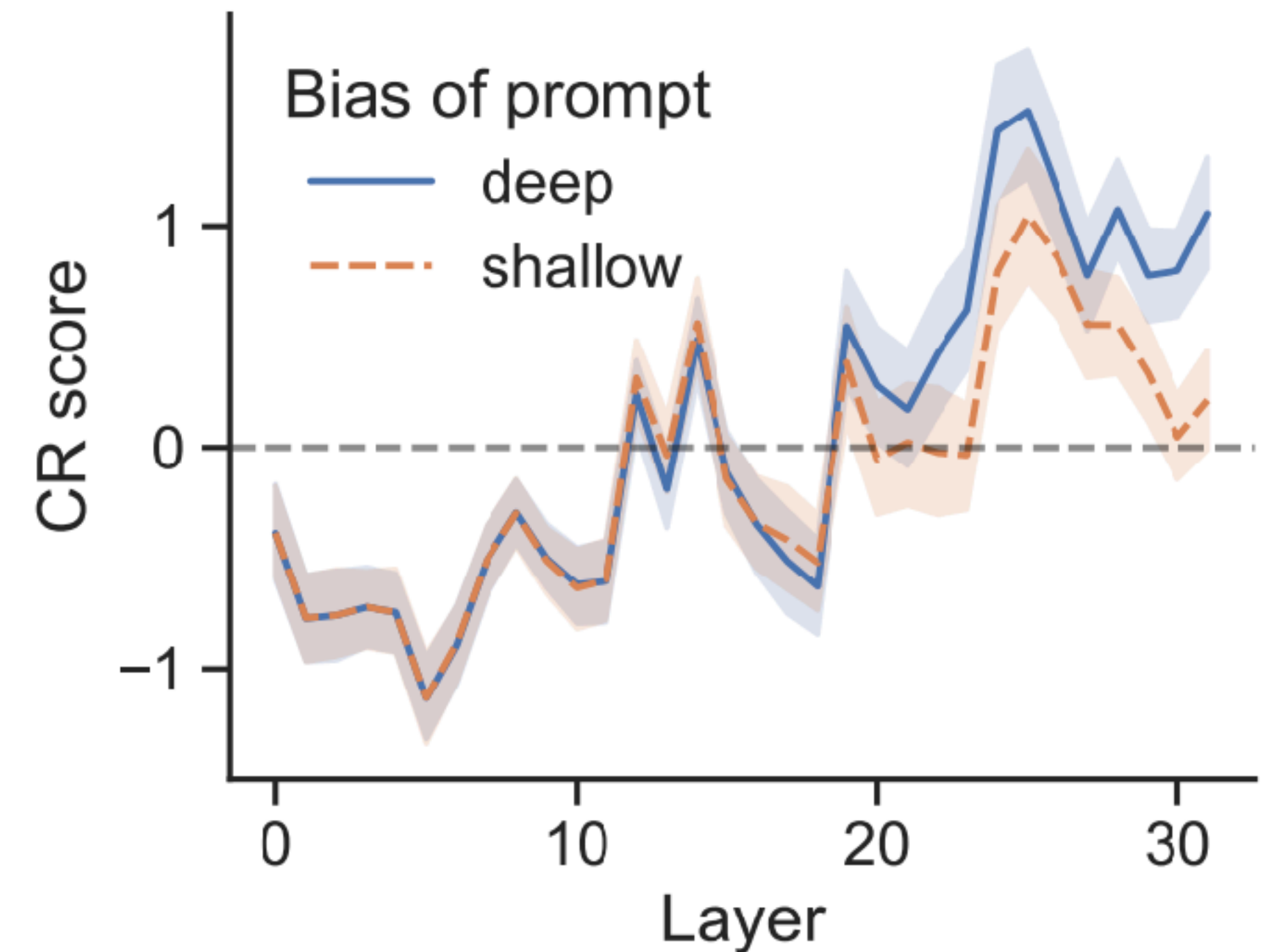


Hagendorff, Fabi & Kosinski (2023, Nature Computational Science)

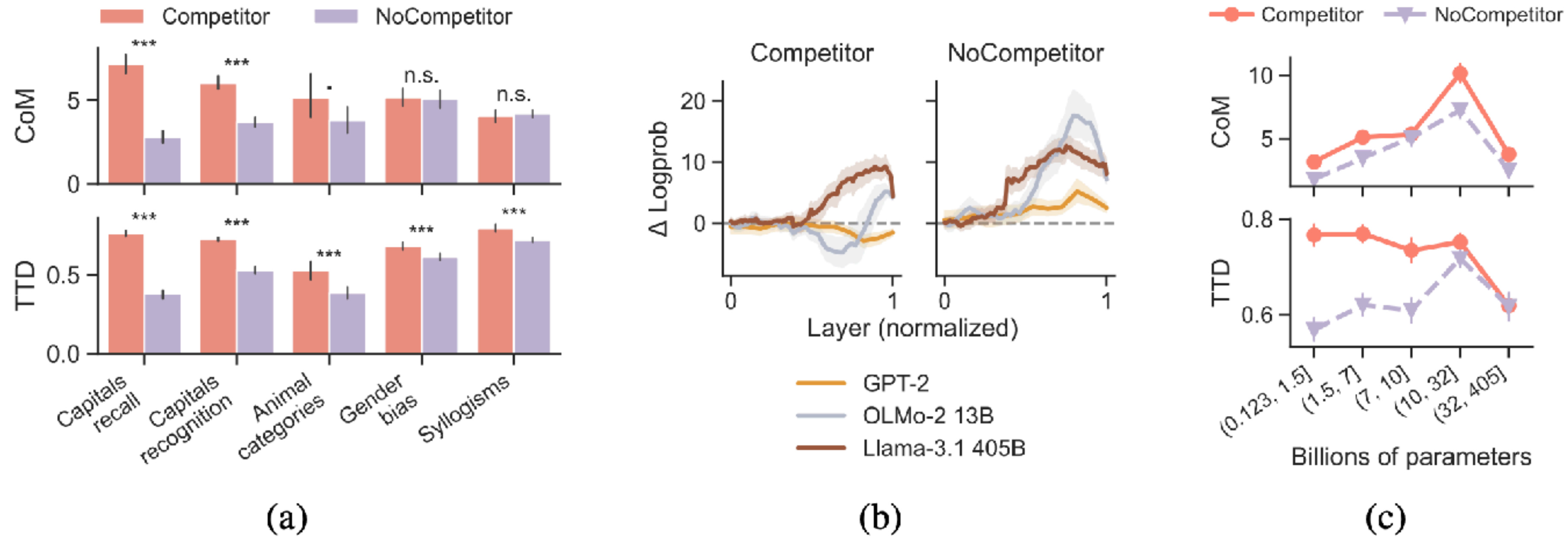


# Deep & shallow thinking in a single forward pass

- ▶ cognitive reflection test items
  - A pear and a fridge together cost \$140. The pear costs \$100 more than the fridge. How much is the fridge?
- ▶ additional context manipulation of factors favoring a deep or shallow processing mode (in humans):
  - e.g., being well-rested vs tired
- ▶ using early decoding / logit lens to investigate the log-odds of correct vs intuitive answer
- ▶ results:
  - preference for shallow answer in early layers
  - effect of contextual manipulation as predicted, but only for some of the LMs tested
    - Lama2 7B  | Mistral 7B  | Starling 7B  | Zephyr 7B 



# Signatures of human-like processing in Transformer forward passes



**Figure 2: Experiment 1 results.** (a) LMs generally show stronger signs of two-stage processing for the items with competing intuitive answers. Asterisks denote sig.  $t$ -tests comparing means across conditions within each domain. (b)  $\Delta \text{LOGPROB}$  across layers for sample LMs in the capitals recall domain, illustrating different processing strategies. (c) Two-stage processing interacts with size.

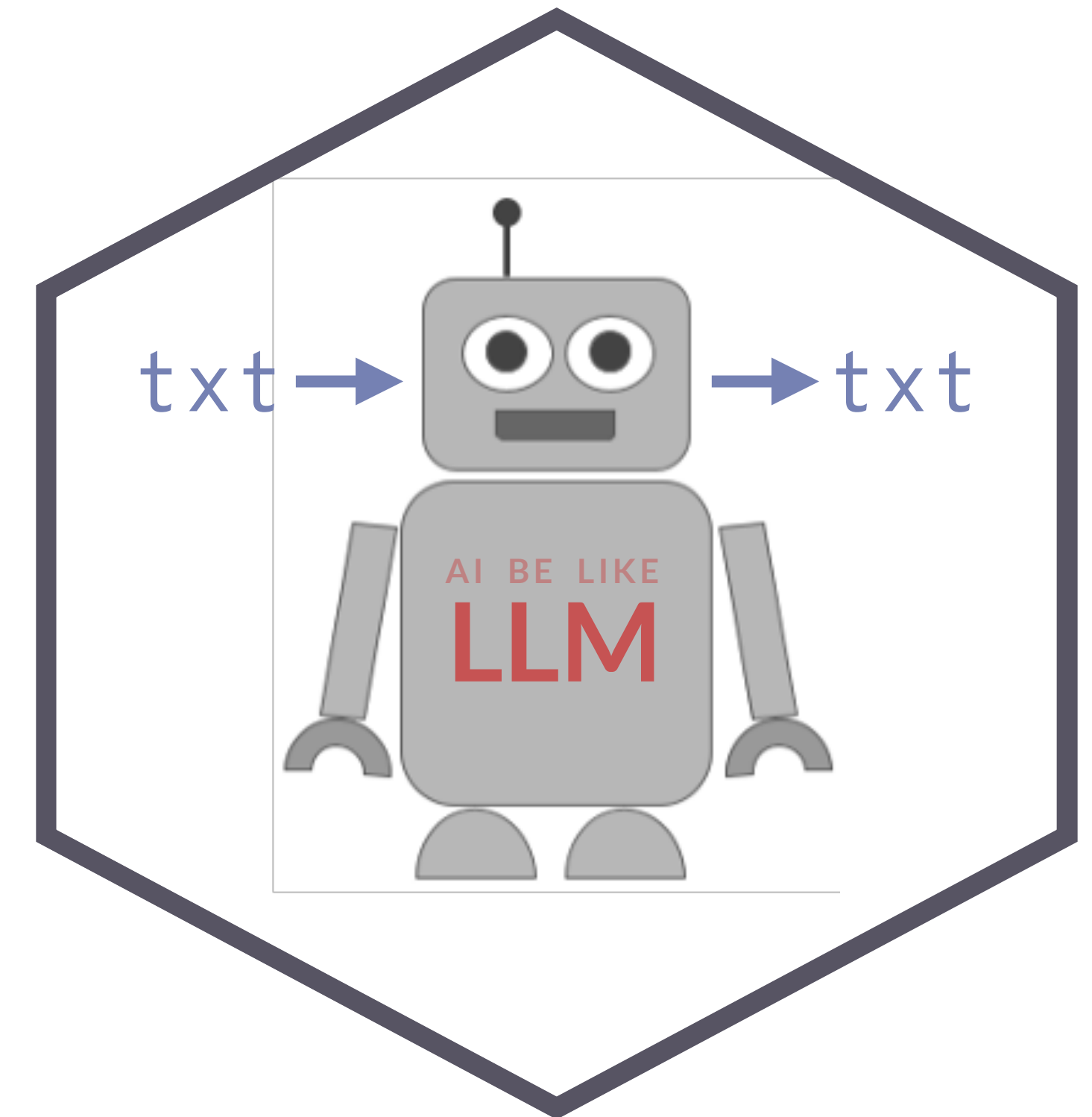
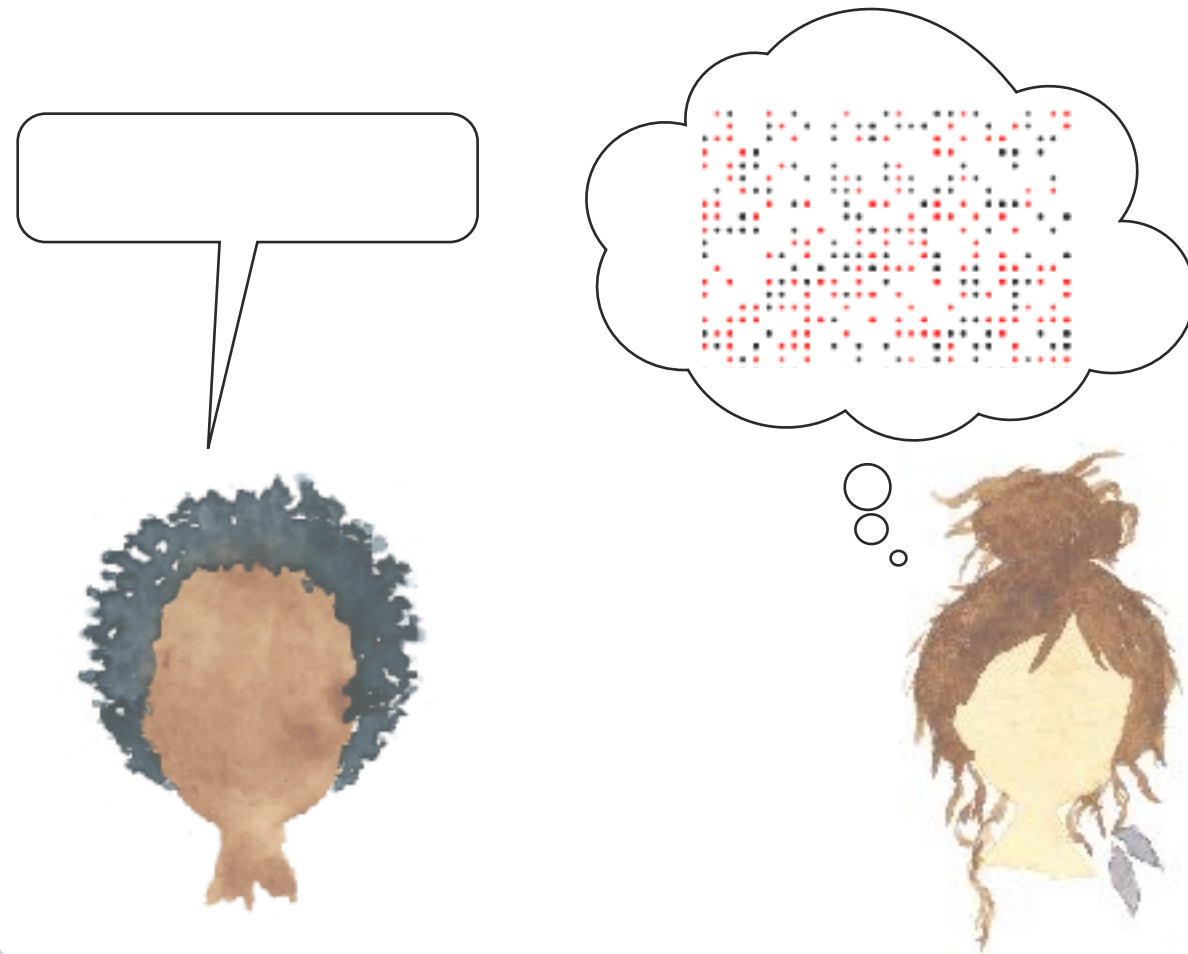


# Targeted assessment of LM's linguistic capacities



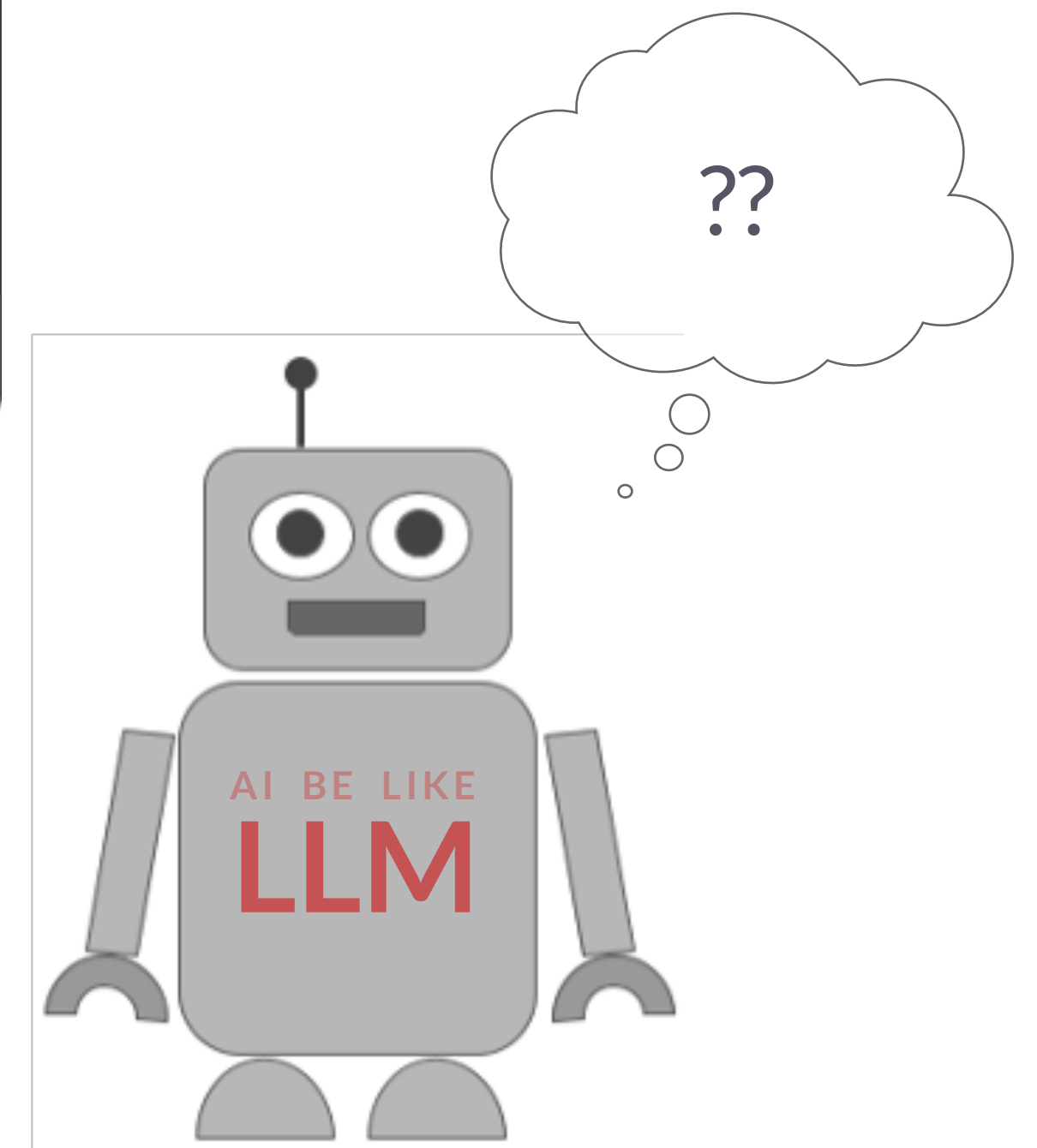
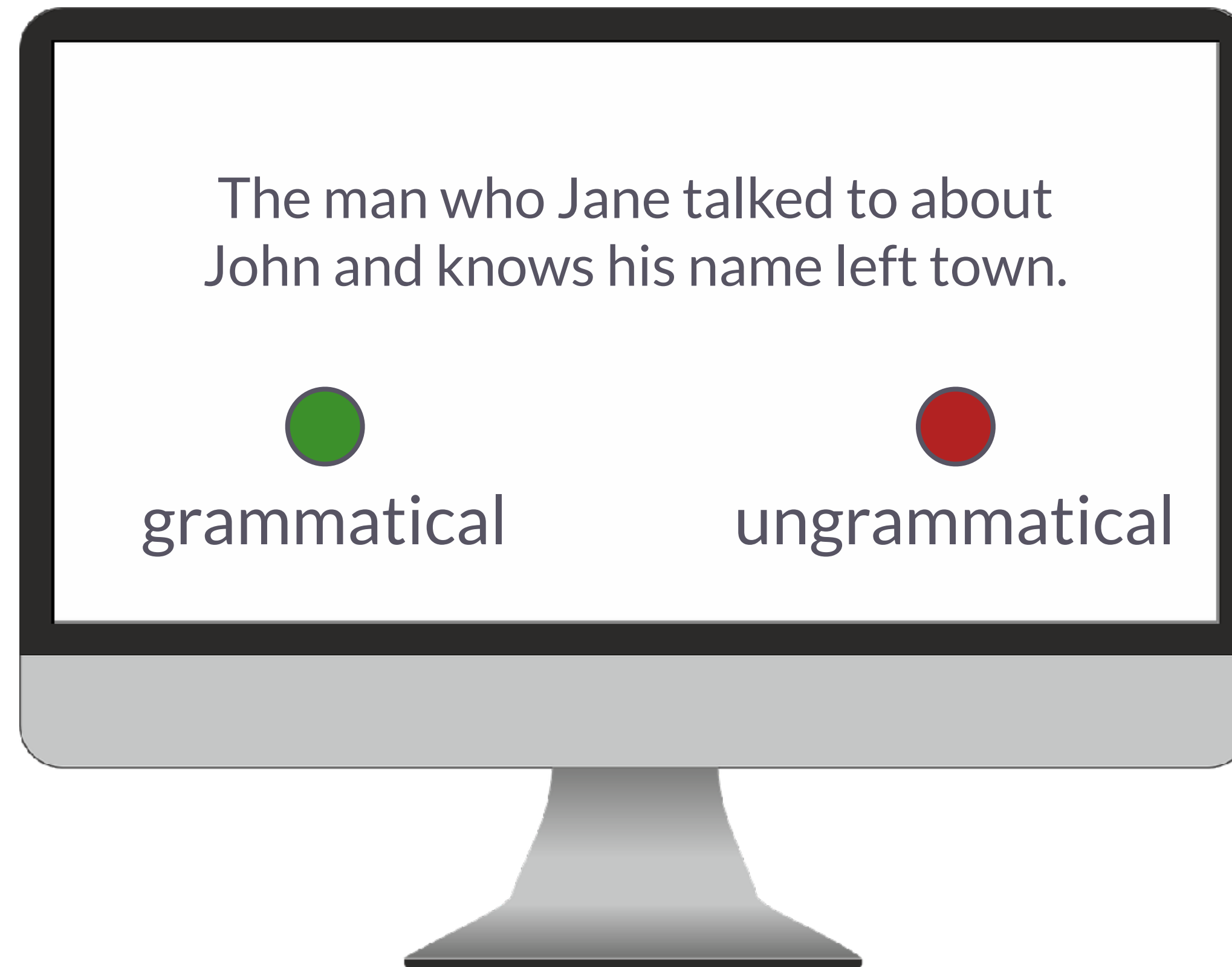
# Language: solved!

language use



# Behavioral experiments

w/ minds & machine



# Targeted behavioral assessment

## ► research question:

- does model  $M$  accurately predict
  - human (offline) grammaticality judgements and/ or
  - human (online) processing data?

## ► method:

- curated test suites (informed by theoretical linguistics & psycholinguistics)
  - e.g., benchmark data set BLiMP (Warstadt et al. 2020)
- derive model predictions from pre-trained models
- compare against armchair judgements or actual human data



# “Targeted Syntactic Evaluation of LMs”

- ▶ **three LMs are compared against each other and human data**
  - n-gram baseline
  - RNN trained on unannotated data
  - same RNN but with additional CCG supertagging
- ▶ **test set: ~350k automatically generated sentence pairs**
  - generated with a non-recursive context-free grammar
- ▶ **focus on three phenomena:**
  - (i) subject-verb agreement, (ii) reflexive anaphora and (iii) negative polarity
- ▶ **main findings:**
  - performance on training data tracks performance of predicting human grammaticality judgements
  - n-gram baseline < simple RNN < multi-trained RNN

## resources

- ▶ [paper](#)
- ▶ [code](#)
- ▶ [video](#)

# Test sentence pairs: SV-Agreement 1

- ▶ **simple agreement**

The author laughs.

\* The author laugh.

The authors laugh.

\* The authors laughs.

- ▶ **agreement in a sentential complement**

The bankers knew the officer smiles.

\* The bankers knew the officer smile.

...

- ▶ **agreement across a prepositional phrase**

The farmers near the parents smile.

\* The farmers near the parents smiles.

...

# Test sentence pairs: SV-Agreement 2

- ▶ **agreement across a subject relative clause**

- \* The officers that love the skater smile.

- \* The officers that love the skater smiles.

- ...

- ▶ **short VP coordination**

- \* The senator smiles and laughs.

- \* The senator smiles and laugh.

- ...

- ▶ **long VP coordination**

- \* The manager writes a letter every day and likes sweets.

- \* The manager writes a letter every day and like sweets.

- ...



# Test sentence pairs: Agreement in object relative clauses

more difficult: model would need to tell two subjects apart

## ► agreement across object relative clauses

The farmer that the parents love swims.

\* The farmer that the parents love swim.

The farmers that the parent loves swim.

\* The farmers that the parent loves swims.

## ► agreement within object relative clauses

The farmer that the parents love swims.

\* The farmer that the parents loves swims.

The farmers that the parent loves swim.

\* The farmers that the parent love swim.

# Test sentence pairs: Agreement in object relative clauses

more difficult: model would need to tell two subjects apart

## ▶ simple reflexive

The senators embarrassed themselves.

\* The senators embarrassed herself.

...

## ▶ reflexive in a sentential complement

The bankers thought the pilot embarrassed herself.

\* The bankers thought the pilot embarrassed themselves.

...

## ▶ reflexive across an object relative clause

The manager that the architects like doubted herself.

\* The manager that the architects like doubted themselves.

...

# Test sentence pairs: Negative polarity

- ▶ **simple NPI**

No students have ever lived here.

\* Most students have ever lived here.

- ▶ **NPI across a relative clause**

No authors the guards like have ever been famous.

\* The authors no guards like have ever been famous.



# Human data

- ▶ 100 participants (MTurk)
- ▶ each participant saw 76 pairs of sentences
- ▶ on each trial, participants had to choose the grammatical sentence from the pair (forced-choice task)
- ▶ 16 participants were excluded due to more than one error on the simple agreement trials



# Think break

1. Given a language model, how would we determine whether the model can or cannot match human grammaticality judgements for any pair of sentences without training the model on the task?
2. If human participants make mistakes, what should we expect an LM to do? Be equally good as humans, or be at ceiling where humans fail to meet the grammatical norm?





# Defining grammaticality prediction

- ▶ given a contrast pair of sentences like:

No students have ever lived here.  $[w_{1:n}]$

\* Most students have ever lived here.  $[v_{1:m}]$

- ▶ an LM is said to predict the right grammaticality judgement iff:

$$P_M(w_{1:n}) > P_M(v_{1:m})$$



# Results

Marvin & Linzen (2018) EMNLP

|                                             | RNN  | Multitask | <i>n</i> -gram | Humans | # sents |
|---------------------------------------------|------|-----------|----------------|--------|---------|
| SUBJECT-VERB AGREEMENT:                     |      |           |                |        |         |
| Simple                                      | 0.94 | 1.00      | 0.79           | 0.96   | 280     |
| In a sentential complement                  | 0.99 | 0.93      | 0.79           | 0.93   | 3360    |
| Short VP coordination                       | 0.90 | 0.90      | 0.51           | 0.94   | 1680    |
| Long VP coordination                        | 0.61 | 0.81      | 0.50           | 0.82   | 800     |
| Across a prepositional phrase               | 0.57 | 0.69      | 0.50           | 0.85   | 44800   |
| Across a subject relative clause            | 0.56 | 0.74      | 0.50           | 0.88   | 22400   |
| Across an object relative clause            | 0.50 | 0.57      | 0.50           | 0.85   | 44800   |
| Across an object relative (no <i>that</i> ) | 0.52 | 0.52      | 0.50           | 0.82   | 44800   |
| In an object relative clause                | 0.84 | 0.89      | 0.50           | 0.78   | 44800   |
| In an object relative (no <i>that</i> )     | 0.71 | 0.81      | 0.50           | 0.79   | 44800   |
| REFLEXIVE ANAPHORA:                         |      |           |                |        |         |
| Simple                                      | 0.83 | 0.86      | 0.50           | 0.96   | 560     |
| In a sentential complement                  | 0.86 | 0.83      | 0.50           | 0.91   | 6720    |
| Across a relative clause                    | 0.55 | 0.56      | 0.50           | 0.87   | 44800   |
| NEGATIVE POLARITY ITEMS:                    |      |           |                |        |         |
| Simple                                      | 0.40 | 0.48      | 0.06           | 0.98   | 792     |
| Across a relative clause                    | 0.41 | 0.73      | 0.60           | 0.81   | 31680   |

# Grammaticality $\neq$ string probability

“The notion ‘grammatical in English’ cannot be identified in any way with the notion ‘high order of statistical approximation to English.’”

Chomsky (1957)

$P(\text{“Snails died the old”} \mid \text{GPT-2}) \approx 2e^{-49}$

$> P(\text{“The ancient crustaceans expired”} \mid \text{GPT-2}) \approx 2e^{-55}$

Wilcox et al. (2023)

# Towards systematic assessment of syntactic generalization

- ▶ 10 LMs are compared against each other, of which 5 non-pretrained:
  - n-gram baseline, vanilla LSTM, ordered neurons LSTM, RNNG, GTP-2
- ▶ 4 different training set sizes (for non-pretrained models)
  - 1, 5, 14 and 42 million tokens
- ▶ test set consists of 34 test suits from 6 “syntactic circuits”
  - (i) garden-path effects, (ii) licensing, (iii) agreement, (iv) center embedding
  - (v) long-distance dependencies, (vi) gross syntactic expectation
- ▶ introduce **syntactic generalization (SG) score**
- ▶ main findings:
  - dissociation between perplexity and SG score
  - model type has more effect on SG than training data size
  - higher SG scores for models with explicit structural training
  - differences in success on different test suits depends on model type

## resources

- ▶ [paper](#)
- ▶ [code](#)
- ▶ [video](#)

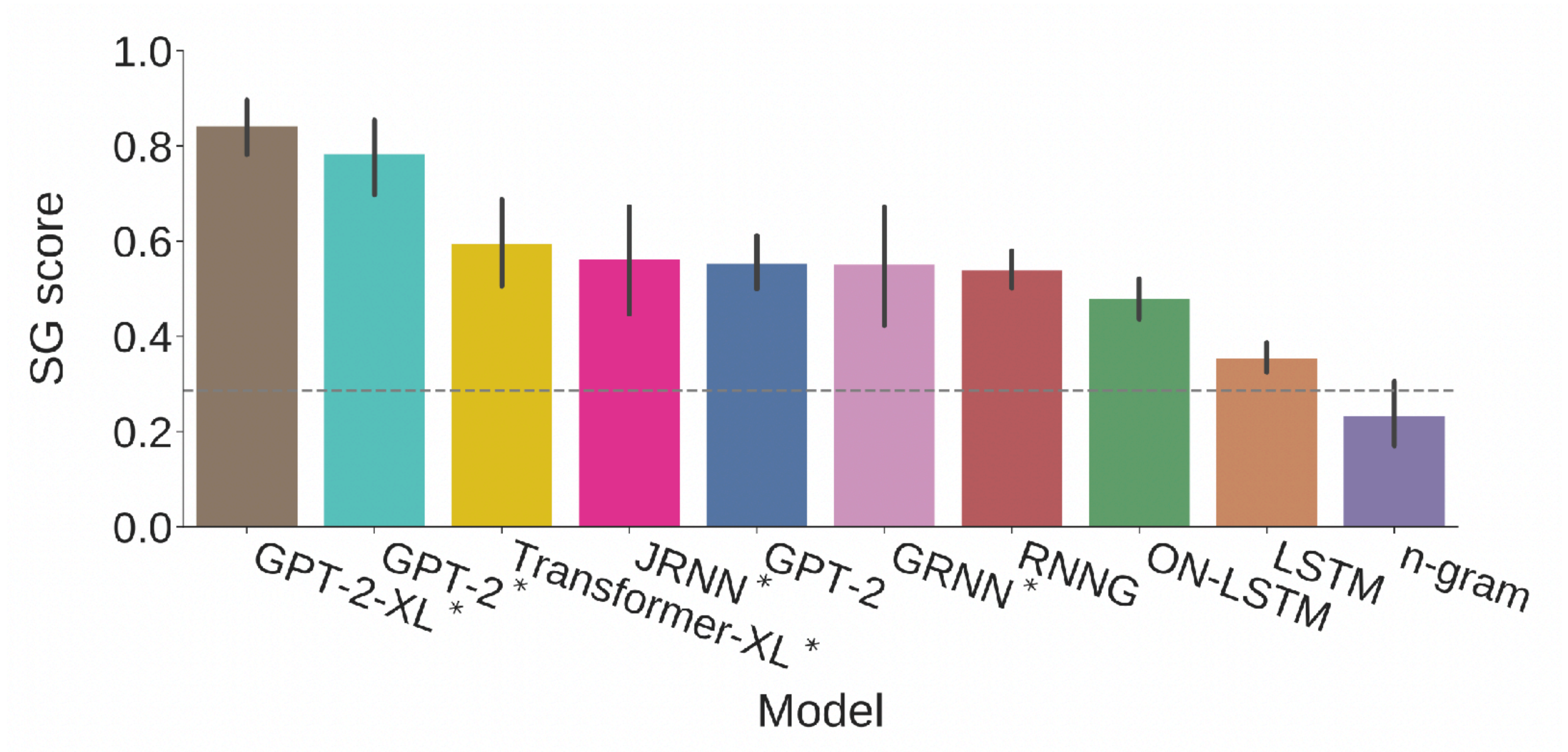


# Syntactic generalization (SG) score

- ▶ each test suit has a set of predictions
- ▶ SG score for test suit X is the proportion of items in X for which the LM matches all predictions associated with X
- ▶ example “garden-path sentences”
  - test item example:
    - i. The horse raced past the barn fell ...
    - ii. The horse ridden past the barn fell ...
    - iii. The horse which was raced past the barn fell ...
    - iv. The horse which was ridden past the barn fell ...
  - associated predictions:
$$P(\text{fell} \mid (i)) < P(\text{fell} \mid (ii))$$
$$P(\text{fell} \mid (i)) < P(\text{fell} \mid (iii))$$
$$P(\text{fell} \mid (i)) - P(\text{fell} \mid (ii)) > P(\text{fell} \mid (iii)) - P(\text{fell} \mid (iv))$$



## Results: Average SG scores by model type





# Taking stock: targeted assessment

of LLM's syntactic abilities

- ▶ careful design to separate string likelihood from grammaticality
  - e.g., test suits & syntactic generalization scores
- ▶ training set size is important, but model architecture (scale) even more







# Assessing language processing

# Sources of processing difficulty

- ▶ limits of working memory

The dog which the cat which the mouse provoked was chased by barked.

- ▶ local ambiguity

The horse raced past the barn fell.

- ▶ interaction w/ semantics & world knowledge

The cop arrested by the detective was guilty of taking bribes.

# Surprisal theory

## ► surprisal theory:

- $\text{Effort}(w_i, w_{1:i-1}, C) \propto \text{Surprisal}(w_i \mid w_{1:i-1}, C) = -\log P(w_i \mid w_{1:i-1}, C)$
- compatible with two mechanisms causing processing difficulty:
  - **prediction**: comprehenders actively predict upcoming words; processing difficulty is a form of prediction error
  - **integration**: comprehenders do not actively predict upcoming material, but passive pre-activation leads to easier integration of some material than others
- empirical evidence for surprisal theory:
  - cloze probability
  - eye-tracked reading
  - self-paced reading
  - EEG during reading
  - maze task



OMG, I didn't see  
that token coming



# Play break

- ▶ go try out the iMaze task for yourself:
  - follow [this link](#)





# Targeted Assessment of Incremental Processing in nLMs & Human

## ► **language models:**

- JRNN: large-scale RNN using LSTM units & CNN character embeddings
- GRNN: from Gulordava et al. (2018)
- GPT-2: version from **1m-zoo** distribution
- RNNG: average of three RNNGs from Hu et al. (2020)

## ► **test set: 16 test suits adapted from Hu et al. (2020)**

## ► **human data on sentence processing difficulty: decision times from an iMaze task**

## resources

- [paper](#)
- [code](#)
- [video](#)

# Targeted Assessment of Incremental Processing in LMs & Human

## ► measure of interest:

- *qualitative*: accuracy scores (LM prediction vs armchair grammaticality judgements)
- *quantitative*: degree of slowdown on critical region (LM prediction vs iMaze data)
- *generalization*: train linear model to map  $P_M(w_i \mid w_{1:i-1}) \mapsto RT_{\text{human}}(w_i \mid w_{1:i-1})$  for each  $w_i$  not in a critical region, and use it to explain RTs from words in critical regions

## ► main findings:

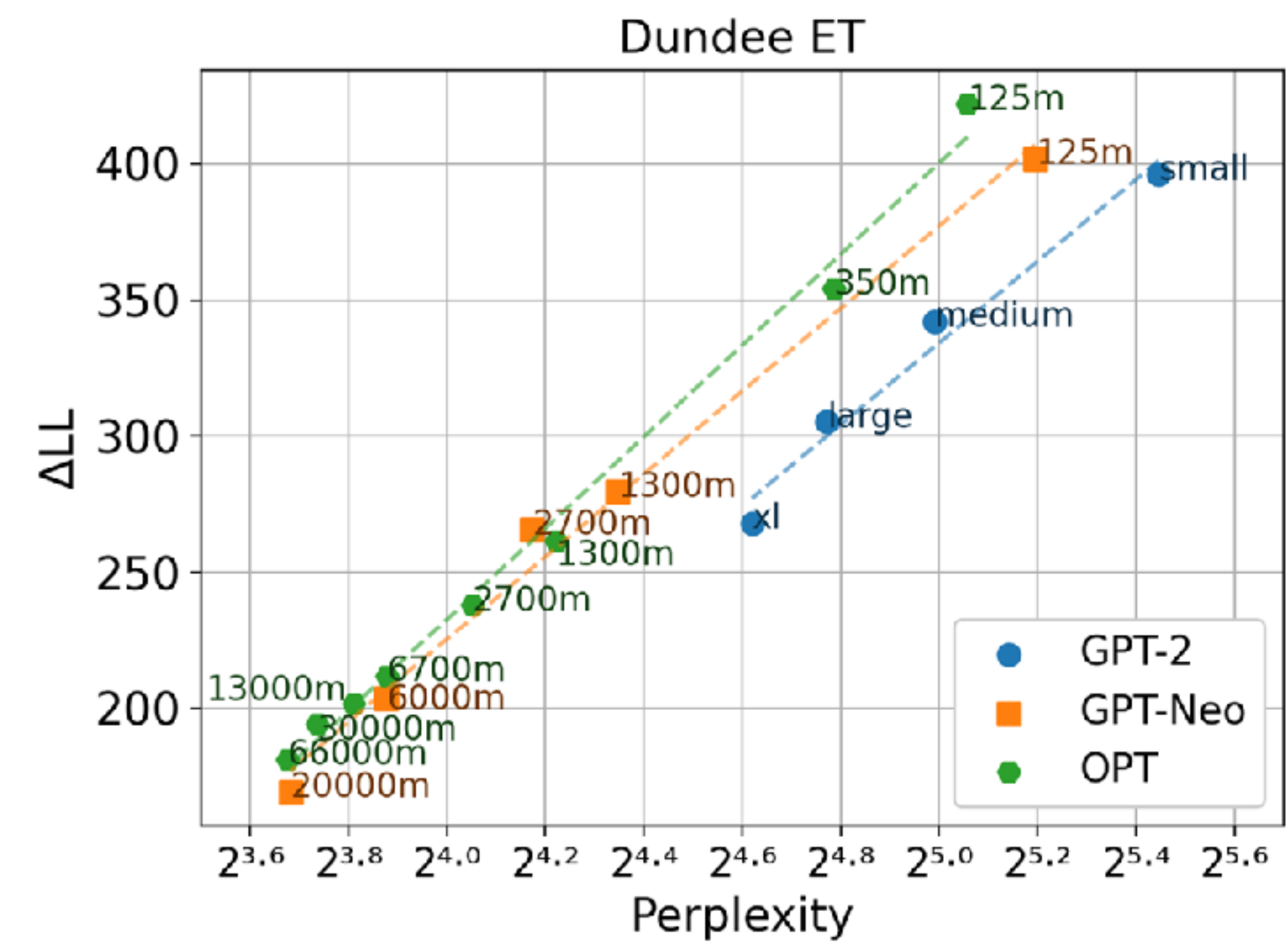
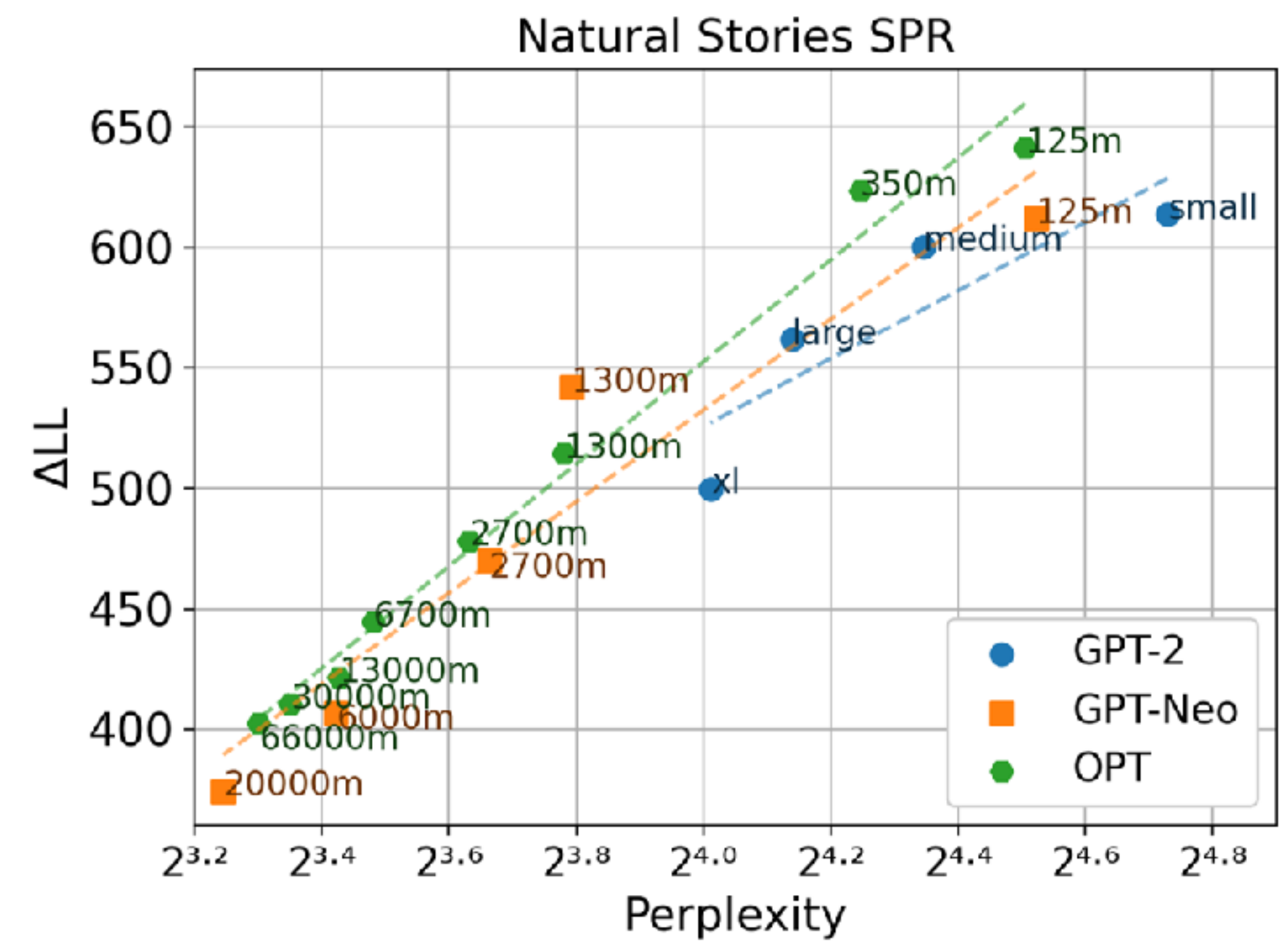
- *qualitative*: LMs predict processing difficulty at regions exactly where humans seem to experience it
- *quantitative*: LMs are "not surprised enough"
- *generalization*: LMs routinely underpredict human RTs / surprisal

## resources

- [paper](#)
- [code](#)
- [video](#)

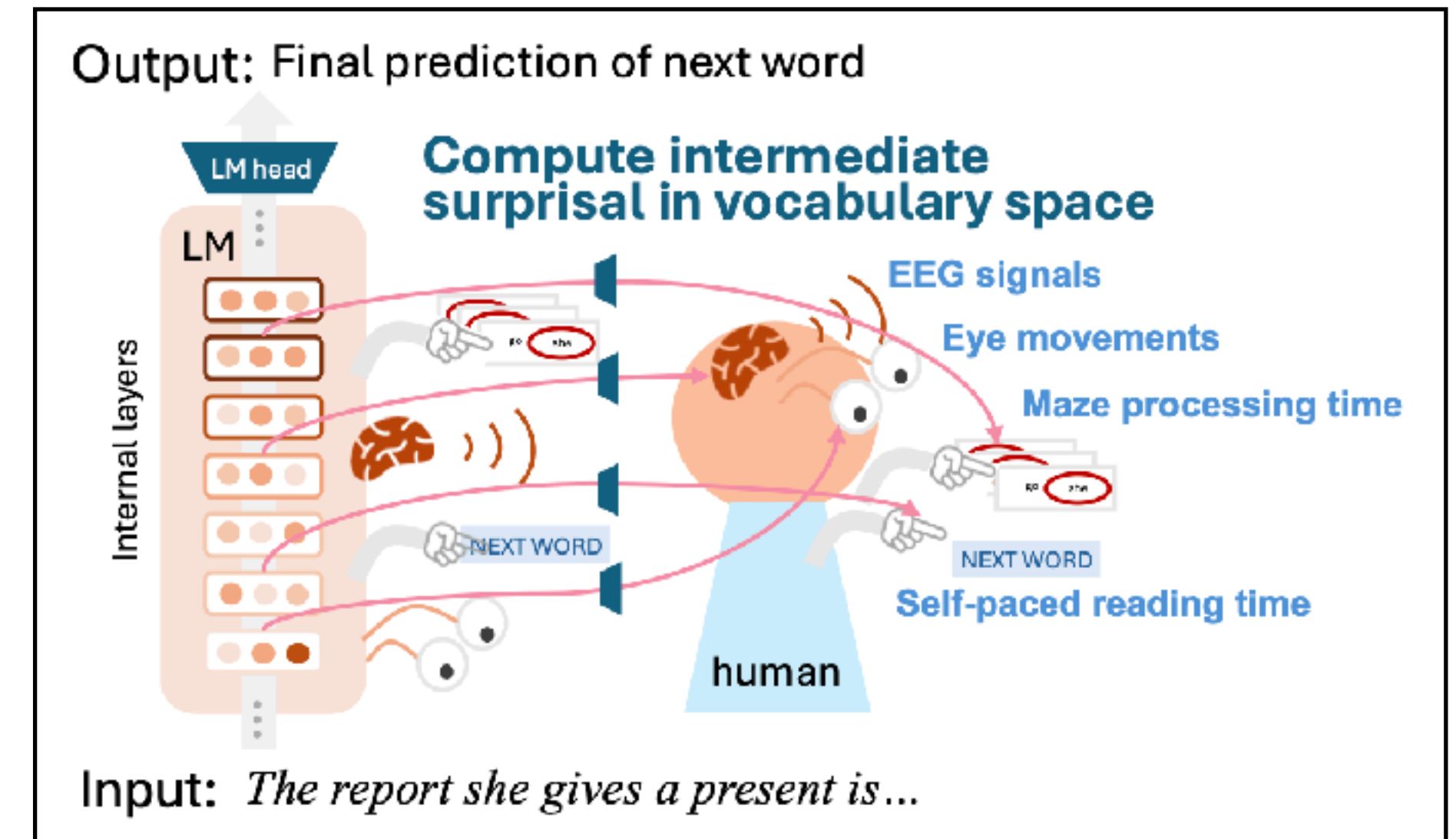


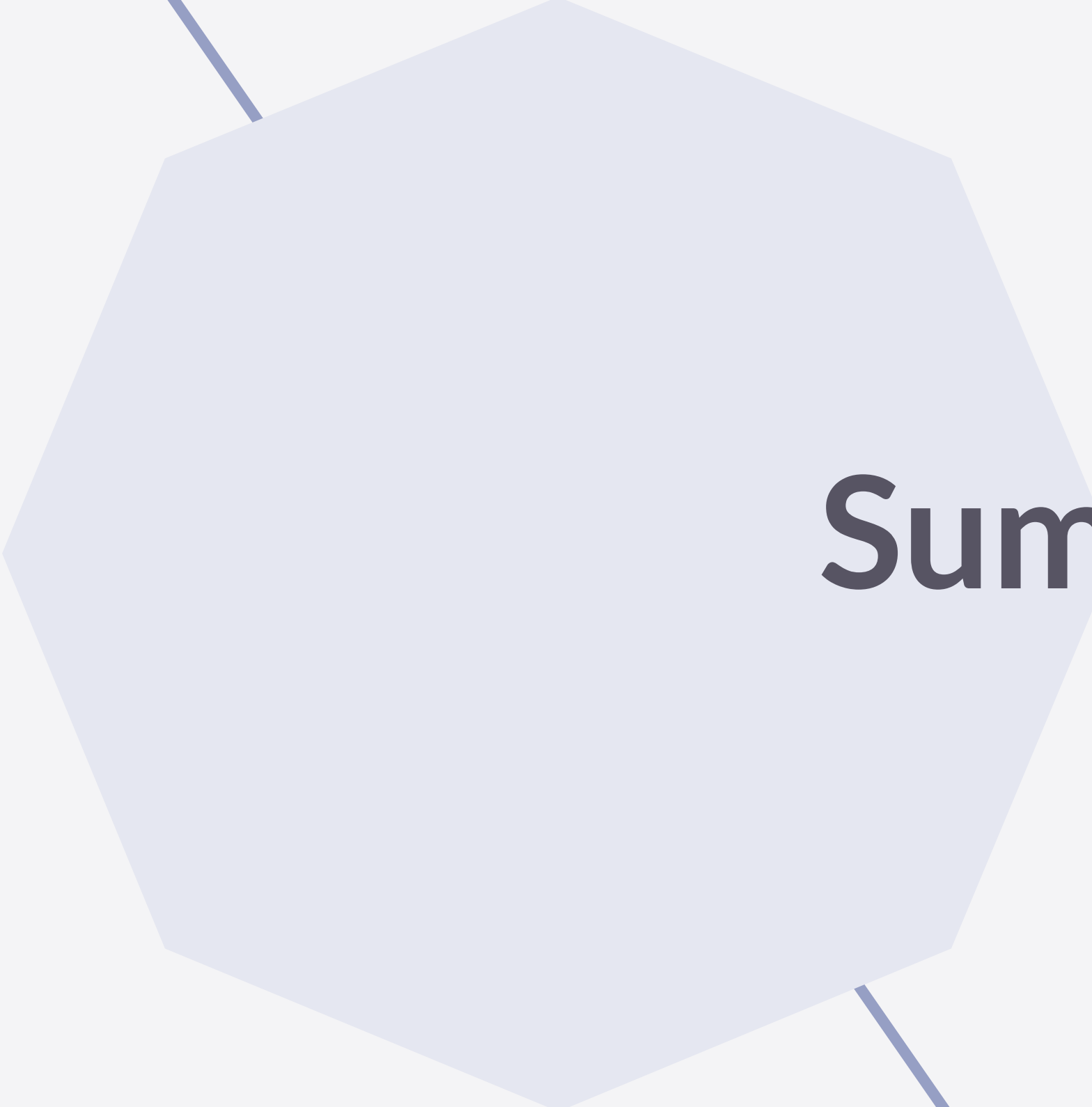
# The larger and “better” the model, the worse it’s predictions



# Early-decoding explains early processing measures

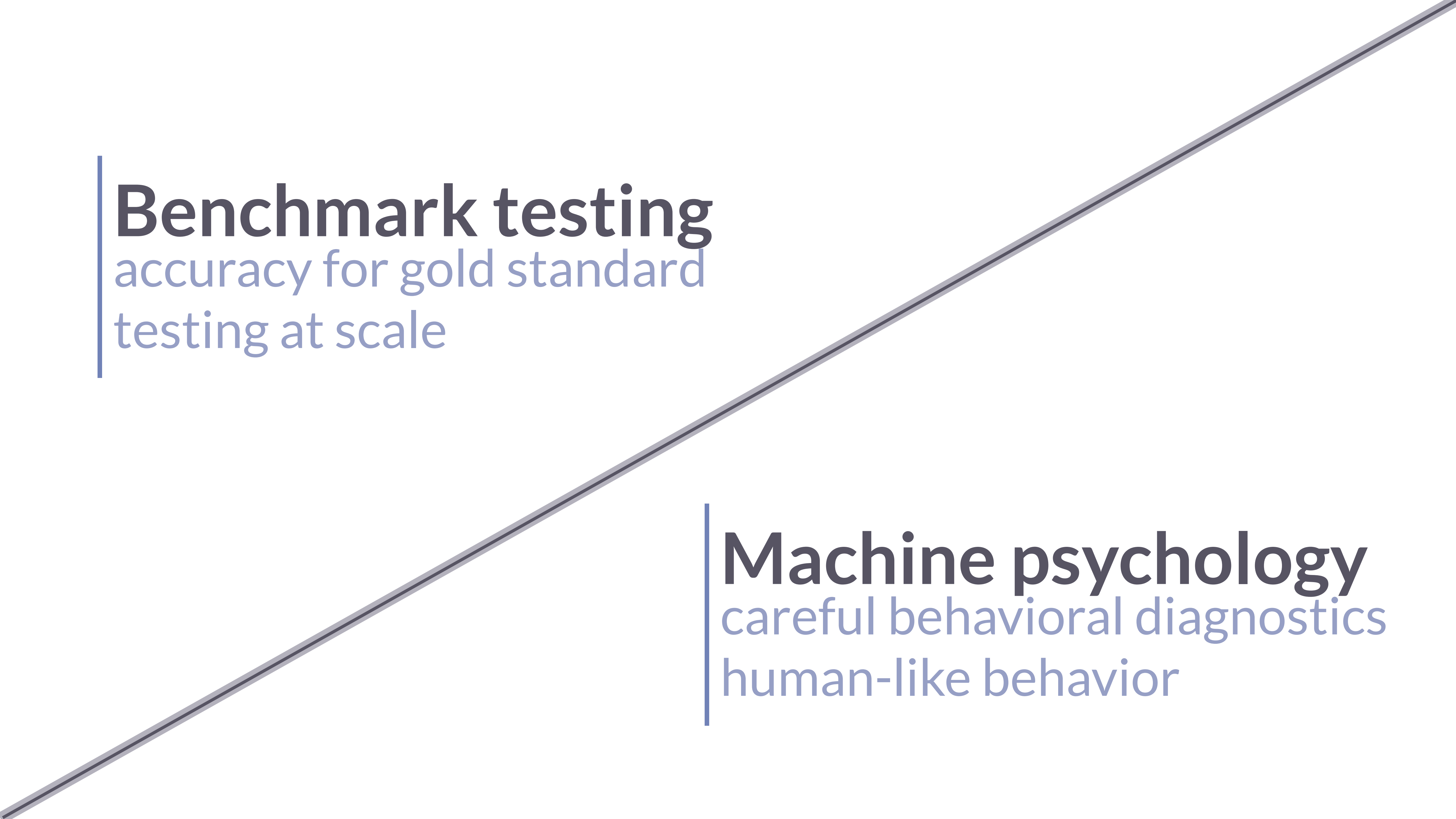
- ▶ better explanations of different behavioral measures from different models sizes or layer-readout locations
- ▶ small models / early readouts  $\leftrightarrow$  “fast measures”
  - first-pace gaze duration
  - self-paced reading times
- ▶ big models / late readouts  $\leftrightarrow$  “slow measures”
  - N400 amplitude
  - MAZE task decision times





# Summary & conclusions





**Benchmark testing**  
accuracy for gold standard  
testing at scale

**Machine psychology**  
careful behavioral diagnostics  
human-like behavior

# Machine psychology: recommended best practices

- ▶ **measurement variability**
  - prompt-sensitivity
    - formulate different instructions
    - alter order of lists etc.
    - create various instances of the same condition
  - aggregation method
- ▶ **known LLM biases:**
  - token frequency
  - biases towards attending most to beginning and end of prompt (Zhao et al. 2021)
- ▶ **clear research question / hypothesis**
  - what kind of claim to I want to make about which kind of model (Lampinen 2023)
    - dumb prompting vs best possible prompting strategy
    - single-prompt LLM or complex prompting w/ CoT ...
- ▶ **severe testing, not confirmation seeking**
  - design to challenge the research hypothesis, not confirm it
  - justify the dependent variable / test measure
  - comparison of model behavior under different types of measures
- ▶ **preregister analyses**
- ▶ **include statistical tests for main (preregistered) hypotheses**
- ▶ **no overselling of results**
- ▶ **training contamination (Hagendorff 2023)**
  - make new material, don't reuse existing ones